College Bus Tracker: Software MVP Specification

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Audience: Development team (mobile, backend, web, QA, DevOps)

Companion document: HW-SPEC-1.0, the hardware specification for the ECE team. This document defines what the software does. HW-SPEC defines what the hardware does.

Authority: This document is the source of truth for all software behaviour. If another document conflicts with it, raise the conflict and update both documents.

0. How to read this document

- MUST: required for MVP launch. SHOULD: required unless a reason is recorded in the decision log. MAY: optional.
- Identifiers: F- features, R- business rules, AC- acceptance criteria, API- endpoints, D- default configuration values (section 35), T- test cases (section 32).
- Storage and APIs use UTC. Screens show Indian Standard Time (IST). Distances are in metres. Speeds are in km/h at every API boundary.
- Values marked default are starting values held in configuration. Change them in configuration, never in code.

1. Product summary

1.1 Purpose

One mobile app for iOS and Android, and one web console for transport staff. Together they show the live position of a fleet of 50 or more college buses to students, faculty, drivers, and transport staff. Each bus has a hardware tracker. The driver's phone is the backup source. The system also covers attendance, emergency alerts, stop requests, notifications, and reports.

1.2 Goals

- G1: Show each bus's position within 15 seconds of its latest valid reading.

- G2: Keep tracking running when the hardware tracker fails, using the driver's phone as the backup source.
- G3: Show every user which source a position comes from.
- G4: Deliver emergency alerts to the right people within 5 seconds.
- G5: Record boardings. Alert faculty when a student boarded the morning trip but not the afternoon trip.
- G6: Give transport staff one console for buses, devices, drivers, routes, users, and reports.
- G7: Meet data protection requirements from day one, including verifiable parental consent for students under 18. The substantive DPDP obligations apply from 14 May 2027, so build to them now.

1.3 Non-goals for the MVP

- Passenger counting or occupancy sensing.
- Fare collection or payments.
- Parent accounts (phase 2).
- A public map that needs no sign-in.
- Turn-by-turn navigation for drivers.
- Multi-college sign-up in the UI. The data model includes college_id from day one.
- Vehicle maintenance management beyond tickets and document expiry reminders.

1.4 Success measures (first 60 days)

- Live position available for at least 99% of active-shift minutes, from either source.
- SOS reaches at least one recipient within 5 seconds, 95th percentile.
- At least 95% of boardings recorded within 2 minutes of departure.
- No cross-role data exposure found in security testing.
- Fewer than one complaint per week about wrong or stale locations.

2. Scope

2.1 In scope (MVP)

Authentication and sessions (F-AUTH). Users, roles, and imports (F-USR). Fleet, devices, and tokens (F-FLT). Routes, stops, and timetables (F-RTE). Assignments (F-ASG). Shifts and trips (F-SHF). Ingestion of hardware and phone data (section 11). Source selection (section 13). Live tracking and ETA (F-TRK). Attendance (F-ATT). SOS (F-SOS). Stop requests (F-STP). Notifications (F-NTF). Admin console (F-ADM). Reports (F-RPT). Privacy requests (F-PRV). Operations, testing, and release (sections 30 to 34).

2.2 Dependencies on the hardware team

The software depends on the telemetry contract (section 11) and the event list (section 12). The hardware team must confirm the GNSS source (module-internal or external) before firmware is frozen. That choice affects the device side only. The API stays the same.

3. Glossary

Term	Meaning
Bus	A physical vehicle, identified by bus number (for example, Bus 01).
Hardware tracker	The device on the bus. Identified by device ID such as HW_001.
Driver phone	The phone running driver mode of the app, used as the backup location source.
Source	Where a position came from: HW (hardware tracker) or PHONE (driver's phone).
Active source	The source the server currently uses for a bus's displayed position.
Mode	AUTO, MANUAL_HW, or MANUAL_PHONE. Set by the driver, overridable by the coordinator.
Shift	One driver's work session on one bus, from start to end.

Term	Meaning
Trip	One scheduled run of a route in one direction: MORNING or AFTERNOON. Created by a shift.
Route	An ordered list of stops with a line geometry.
Stop	A named point with a location and a radius.
Roster	The drivers allowed to run a given bus.
Expected passenger	A student assigned to a route for a trip type on a date, and not on leave.
Boarding	A recorded check-in of a student to a trip.
Absence alert	Alert to faculty when a student boarded the morning trip but not the afternoon trip.
SOS	An emergency incident.
Escalation tier	The level of recipients for an unacknowledged SOS.
Device token	Secret credential that a hardware tracker uses to authenticate.
Session	A signed-in login on one device.
Snapshot	One API response containing the state of every bus on a route.
DLT	Distributed Ledger Technology registration. TRAI requires it for SMS sent in India.
Live, Stale, Not running, Unavailable	Bus states defined in section 14.

4. Users and roles

Code	Role	Who	Account created by	Sign-in	Admin MFA	Home screen
STU	Student	Enrolled students	Imported from college records	College SSO	No	Live map
FAC	Faculty	Teachers linked to buses	Coordinator	College SSO	No	My buses

Code	Role	Who	Account created by	Sign-in	Admin MFA	Home screen
DRV	Driver	Bus drivers	Coordinator	Email OTP	No	Start shift
SEC	Security	Gate and campus security	Coordinator	College SSO	Yes	Live map (all active buses)
COR	Transport coordinator	Transport office staff	Super admin	College SSO	Yes	Dashboard
TEC	Technician	Maintenance staff	Coordinator	College SSO	Yes	Device health
MGT	Management	Leadership	Super admin	College SSO	Yes	Aggregate dashboard
SAD	Super admin	Named IT owners (minimum 2)	Set at deployment	College SSO	Yes	Settings

Rules:

- A user may hold several roles. The app shows a role switcher when there is more than one.
- Students and staff sign in with a college account. Drivers may use a driver email if no college account exists.
- At least two super admins must exist at all times.
- Deactivated users disappear from all live views within 60 seconds.
- Every role change is written to the audit log.

5. Permissions matrix

Y = full access. A = assigned buses or own scope only. O = own records only. R = read-only. - = no access.

Capability	STU	FAC	DRV	SEC	COR	TEC	MGT	SAD
View live bus positions	A	A	A	Y (all active)	Y	R	R (aggregate)	Y
See active source label	Y	Y	A	Y	Y	Y	R	Y
See source-switch log	-	-	A	-	Y	Y	R	Y

Capability	STU	FAC	DRV	SEC	COR	TEC	MGT	SAD
Start and end own shift	-	-	Y	-	-	-	-	-
Change mode	-	-	A	-	Y (override)	-	-	Y
Show boarding QR	-	-	A	-	-	-	-	-
Scan QR to board	O	-	-	-	-	-	-	-
Manual check-in	-	-	A	-	Y	-	-	Y
View attendance	O	A	A	-	Y	-	R (aggregate)	Y
Mark leave	O	A	-	-	Y	-	-	Y
Send SOS	Y	Y	Y	Y	Y	-	-	Y
Receive SOS	-	A	A (own bus)	Y	Y	-	-	Y
Acknowledge and resolve SOS	-	A	-	Y	Y	-	-	Y
Submit stop request	O	-	-	-	-	-	-	-
Review stop requests	-	-	-	-	Y	-	-	Y
Manage buses, routes, stops, timetables	-	-	-	-	Y	-	-	Y
Register devices, rotate tokens	-	-	-	-	-	Y	-	Y
Link device to bus	-	-	-	-	Y	-	-	Y
Revoke device token	-	-	-	-	Y	-	-	Y
Manage assignments (roster, faculty, student routes)	-	-	-	-	Y	-	-	Y
Import users, deactivate users	-	-	-	-	Y	-	-	Y
Assign roles	-	-	-	-	-	-	-	Y
Report a bus problem (ticket)	-	-	Y	-	Y	Y	-	Y

Capability	STU	FAC	DRV	SEC	COR	TEC	MGT	SAD
Publish announcements	-	-	-	-	Y	-	-	Y
Reports	O (own attendance)	A	-	-	Y	R (device reports)	R	Y
Audit log	-	-	-	-	R	-	-	Y
Thresholds and feature flags	-	-	-	-	R	-	-	Y
Process personal data requests	O (own)	O (own)	O (own)	O (own)	Y	-	-	Y

Decision point: whether SEC sees all active buses (default: yes, read-only). Confirm before build.

6. Architecture

6.1 Context diagram

```
[Hardware tracker]  --HTTPS, device token------>\
[Driver phone]      --HTTPS, driver session---->  |--> [API service, 2+
containers] --> [PostgreSQL + PostGIS]
[Mobile app]        --HTTPS, user session------->  |        |
(primary + standby)
[Admin web console] --HTTPS, cookie + CSRF----->/        |
                                                         +--> [Push: FCM /
APNs]
[Job workers, 2+] ---> [PostgreSQL job queue] ----------+--> [SMS via DLT
provider]
                                                         +--> [Email: SES]
                                                         +--> [Object
storage: exports]
[Monitoring: logs, metrics, traces, error tracking, external uptime checks]
```

6.2 Components

Component	Responsibility	Scaling	Failure behaviour
API service	Authentication, all reads and writes, device and driver ingestion, snapshot	Stateless. Minimum 2 tasks across 2 availability zones	Load balancer removes unhealthy tasks. Other task keeps serving.
Ingestion routes (inside API)	Validate device and driver messages, store samples, trigger source selection	Same as API. Rate limited	Device retries. Nothing is acknowledged before it is stored.
Source selector	Decides the active source for each bus	Job per bus plus an on-ingest trigger	If it stalls, displayed age keeps rising and an alert fires
Snapshot service	Serves the state of all buses on a route	Cache of 3 seconds per route, per instance	Serves last cached result, marked with its age
Job workers	Background jobs (section 25)	Minimum 2 workers, leased jobs	Retry with backoff. Dead-letter after 6 attempts
Notification dispatcher	Sends push, SMS, and email	Worker pool	Retries (section 21). Failures logged per recipient
Admin web console	Coordinator, technician, admin, and management screens	Static hosting plus API	Read-only banner if API is unreachable
Mobile app	Student, faculty, and driver screens. Driver background location	On device	Cached last-known data with clear banners
PostgreSQL	System of record and job queue	Primary and standby across availability zones	Automatic failover. Target recovery point of 15 minutes or less
Object storage	Export files	Managed	Versioned and private

Component	Responsibility	Scaling	Failure behaviour
Monitoring	Logs, metrics, traces, alerts	Managed	Watchdog alert if alerting itself stops

6.3 Core data flows

- F1 Hardware location: device POSTs samples, then the API validates them (section 11.5), stores them with source HW, and triggers the selector. The selector updates bus_state. The next snapshot shows the new position.
- F2 Phone location: the same path with source PHONE. Accepted only during an active shift for the bus on the driver's roster (R-16).
- F3 SOS: any trigger creates or joins an incident. Fan-out jobs send push and SMS. Acknowledgements stop escalation. Resolution closes the incident (section 19).
- F4 Admin change: the API writes the change and an audit row, then invalidates the route's snapshot cache. Clients see the change on their next poll.
- F5 Boarding: a student scans the QR code, the API validates it, and writes an attendance row. At the cutoff, a job evaluates the afternoon absence rule (section 18).

7. Technology stack

Layer	Decision	Reason	Alternatives considered
Mobile app	React Native with Expo, TypeScript. One codebase for iOS and Android	One codebase serves three roles. Mature background-location libraries. Over-the-air JavaScript updates for non-native fixes	Native Swift and Kotlin (two codebases, higher cost). Flutter (viable; choose on team skills). PWA (weak background tracking and push on iOS)

Layer	Decision	Reason	Alternatives considered
Location (driver)	expo-location with a background task. Android foreground service with location type. iOS Always permission with background location mode	Required by both platforms for background tracking (section 16.3)	Native modules only (more work)
Push	Firebase Cloud Messaging for Android and iOS. APNs key configured in Firebase	One provider for both platforms	OneSignal (adds a vendor)
Map	MapLibre React Native with OpenStreetMap tiles	No per-load charges. Works in the app and the web console	Google Maps Platform (usage charges; can be swapped in later if budget allows)
Admin web	React with Vite, TypeScript. shadcn/ui components. TanStack Query. React Hook Form	Fast, accessible, common skills	Next.js (not needed for an internal console)
Shared contracts	Zod schemas in packages/contracts, used by API, mobile, and web	One definition for validation and types	OpenAPI-first with generated clients
API	Node.js LTS, TypeScript, NestJS. OpenAPI 3.1 generated from code	Module structure suits a team. Built-in validation hooks	Fastify (lighter). Express (less structure)
ORM and migrations	Prisma for schema and migrations. Raw SQL for PostGIS queries	Safe migrations. PostGIS needs SQL	Drizzle ORM. TypeORM

Layer	Decision	Reason	Alternatives considered
Database	PostgreSQL 16 or later with PostGIS, managed, Multi-AZ	Dated links, transactions for SOS, geographic queries for route corridors and ETA	MongoDB Atlas (the earlier plan). Workable, but geographic queries and dated relationships are weaker. See section 41
Job queue	pg-boss on PostgreSQL	No Redis needed for MVP. Fewer moving parts	BullMQ with Redis (phase 2, for push fan-out)
Cache	In-process cache, 3-second TTL, per instance	Simple. Staleness is acceptable at 3 seconds	Redis (phase 2)
Live updates	Snapshot polled every 10 seconds in the foreground	Lowest complexity. Works across multiple instances	Server-Sent Events or WebSockets with pub/sub (phase 2)
Authentication	College SSO through OpenID Connect (Google Workspace or Microsoft Entra ID). Email OTP for drivers. The app issues its own short-lived tokens	No password storage	Auth0, Firebase Auth, Amazon Cognito
SMS	DLT-registered transactional SMS provider with approved sender ID and templates	TRAI requirement	None
Email	Amazon SES	Low cost, reliable	None
File storage	Amazon S3. Private, versioned, signed URLs valid for 24 hours	Standard	None

Layer	Decision	Reason	Alternatives considered
Hosting	AWS: ECS on Fargate (minimum 2 tasks, 2 zones), Application Load Balancer, RDS PostgreSQL Multi-AZ, S3, SES, Secrets Manager, KMS, CloudWatch, WAF, ACM certificates	Managed services. Meets the reliability plan (two instances, managed database with standby)	Google Cloud Run with Cloud SQL (equivalent). Serverless functions (considered; long-running jobs and future push favour containers)
Infrastructure as code	Terraform	Repeatable environments	CloudFormation
Monitoring	CloudWatch metrics and logs. Sentry for errors. OpenTelemetry tracing with sampling. External uptime checks	Standard. Low operating effort	Datadog, Grafana Cloud
CI/CD	GitHub Actions. EAS Build and EAS Submit for mobile	Standard tooling	GitLab CI
Testing	Vitest (unit), Supertest with Testcontainers (API integration), Playwright (web end-to-end), Maestro (mobile end-to-end), k6 (load), OWASP ZAP (security scan)	Open source and well known	None
Repository	pnpm workspaces with Turborepo	One repository for apps and services	Separate repositories
Code quality	TypeScript strict mode, ESLint, Prettier, pre-commit hooks, conventional commits	Consistency	None

Pin exact versions when the repository is created. Use the current LTS release of each runtime on that day, and record the versions in the repository's README.

7.1 Repository layout

```
bus-tracker/
  apps/mobile          React Native (Expo) app: student, faculty, driver
  apps/admin-web       React admin console: coordinator, technician, admin,
management, security
  services/api         NestJS API: REST, ingestion, auth, OpenAPI
  services/worker      Same codebase, job runner process
  packages/contracts   Zod schemas shared by API, mobile, and web
  packages/db          Prisma schema, migrations, PostGIS helpers
  tools/simulator      Device and phone simulators (local, staging only)
  infra/               Terraform for all environments
  docs/                This specification, API reference, runbooks
```

7.2 Environments

Environment	Purpose	Data	Deployment
Local	Development. Docker Compose with PostgreSQL and PostGIS, MinIO (S3-compatible), Mailpit (email)	Synthetic	Developer machine
Dev	Shared integration. Latest main branch	Synthetic	Automatic on merge
Staging	Production-like. Simulated fleet of 50 buses and 2,000 accounts. Connected to a hardware test bench	Synthetic only	Automatic on main. Approval for production-like changes
Production	Live service	Real	Manual approval

Rule: real student, driver, or location data exists only in production. Staging and development use synthetic data only.

8. Identity, authentication, and sessions

- F-AUTH-01 College SSO. Sign-in uses OpenID Connect. Only accounts from the college domain are accepted. On first sign-in, the system matches the email to an imported user record. If there is no match, the user sees: contact the transport office.

- F-AUTH-02 Driver email OTP. A 6-digit code, valid for 10 minutes. Five attempts per code. Five codes per hour per account. Ten consecutive failures lock the account for 15 minutes.
- F-AUTH-03 Sessions. Each sign-in creates a session family. Access token: signed JWT, 15 minutes. Refresh token: opaque, 30 days for students and staff, 12 hours for drivers. Refresh tokens rotate on every use. Reuse of an old refresh token revokes the whole family and forces sign-in.
- F-AUTH-04 Admin MFA. TOTP enrolment at first sign-in. Required for SEC, COR, TEC, MGT, and SAD. Ten single-use recovery codes, shown once.
- F-AUTH-05 Session management. Profile screen lists active sessions and offers sign-out of other devices.
- F-AUTH-06 Deactivation. User status is checked on every request, with a cache of at most 30 seconds. All sessions are revoked on deactivation.
- F-AUTH-07 No passwords are stored or accepted in the MVP.

Client storage rules:

- Mobile: tokens are kept only in platform secure storage (Expo SecureStore).
- Web: tokens are kept in httpOnly, Secure, SameSite=Strict cookies. State-changing requests require a CSRF token header.
- Tokens are never written to local storage, logs, analytics, or crash reports.

Driver session scope: a driver session can write location only while a shift is active for a bus on that driver's roster (R-16). The session token also carries the shift_id, so a token cannot be reused for another shift.

Device authentication is defined in section 11.1.

9. Domain model and business rules

9.1 Relationships

```
College 1--* User
User *--* Role                          (user_roles, dated)
Bus 1--* DeviceAssignment *--1 Device   (hardware link, dated)
Bus 1--* RosterEntry *--1 User(DRV)     (dated)
Bus *--* User(FAC)                      (faculty_assignments, dated, primary
flag)
Route 1--* RouteStop *--1 Stop          (ordered)
Route *--* Bus via RouteBusAssignment   (trip type, weekday mask, dated)
Route 1--* Timetable                    (departure times per trip type)
User(STU) *--1 Route via StudentRoute   (trip type, boarding stop, dated)
Shift *--1 Bus, *--1 User(DRV), 1--1 Trip
Trip 1--* AttendanceRecord *--1 User(STU)
Bus 1--1 BusState
Bus 1--* LocationSample                 (source HW or PHONE)
SosIncident 1--* SosEvent, *--1 Bus
StopRequest *--1 Route
```

9.2 Business rules

- R-01: A bus has at most one active hardware link at a time.
- R-02: A device has at most one active bus link at a time.
- R-03: A driver may be rostered on several buses but may run only one shift at a time.
- R-04: A bus may run only one shift at a time.
- R-05: Links, rosters, and assignments are dated. Old rows are closed, never deleted.
- R-06: A student has at most one active route per trip type on a given date.
- R-07: Every sample keeps the device or driver that produced it, even after a reassignment.
- R-08: Records are deactivated, not deleted. Hard deletion happens only through a personal data deletion request or the retention job.
- R-09: Audit log rows are append-only. The application database role has no UPDATE or DELETE permission on the audit table.
- R-10: Only the server chooses the active source. Clients display what the server returns.
- R-11: SOS cannot be disabled by settings, quiet hours, or muting.

- R-12: Students see only buses serving their route on today's active trips.
- R-13: A stop request never edits an existing stop. Approval creates a new stop.
- R-14: One attendance record per trip and student.
- R-15: All timestamps are stored in UTC.
- R-16: A driver's phone location is accepted only during an active shift, only for the bus on that driver's roster, and only for that bus.
- R-17: Location history is visible only to COR and SAD. Other roles see live and last-known positions only.
- R-18: Phone location is collected only during an active shift and only with the driver's permission.
- R-19: Student devices do not send location to the server in the MVP.
- R-20: A bus with no active shift is shown as Not running, never as Offline.

10. Data model

All tables have id (UUID), created_at, and updated_at unless noted. Enumerations are stored as text with check constraints.

10.1 Identity and access

- users: college_id, email (unique per college), full_name, phone, status (ACTIVE, SUSPENDED, DEACTIVATED), deactivated_at.
- user_roles: user_id, role, granted_by, granted_at, revoked_at.
- sessions: user_id, family_id, refresh_token_hash, device_label, last_used_at, revoked_at.
- mfa_factors: user_id, type (TOTP), secret (encrypted at rest), enrolled_at.
- recovery_codes: user_id, code_hash, used_at.
- otp_challenges: user_id, code_hash, expires_at, attempts, consumed_at.

10.2 Fleet and devices

- buses: bus_number (unique), registration_number (unique), capacity, status (ACTIVE, MAINTENANCE, RETIRED), insurance_expiry, fitness_expiry, pollution_expiry, permit_expiry.

- devices: device_id (unique, format HW_nnn or HW_nnnn), serial_number, imei, iccid, status (PROVISIONED, ACTIVE, REVOKED, RETIRED), fw_version, schema_version, provisioned_at, retired_at.
- device_tokens: device_id, token_hash (HMAC-SHA256 with a server-held key), status (PENDING, ACTIVE, REVOKED), first_used_at, revoked_at, expires_at (for PENDING only).
- device_assignments: bus_id, device_id, valid_from, valid_to (null means current), set_by.
- firmware_releases: version, sha256, size_bytes, storage_key, channel (STABLE, BETA), status (DRAFT, ROLLING_OUT, COMPLETE, WITHDRAWN), released_at.
- device_heartbeats: device_id, boot_id, received_at, fw_version, uptime_s, network_status, sim_status, signal_rssi, vehicle_battery_mv, device_battery_mv, gps_status, buffer_bytes. Partitioned by day.
- maintenance_tickets: bus_id, device_id, source (DRIVER_REPORT, SYSTEM_ALERT, MANUAL), category, status (OPEN, IN_PROGRESS, RESOLVED, CLOSED), opened_by, assigned_to, description, opened_at, closed_at.

10.3 Routes and stops

- stops: name, location (geography Point), radius_m (default 50), status, created_by.
- routes: code (unique), name, status.
- route_stops: route_id, stop_id, sequence (unique per route), valid_from, valid_to.
- route_geometries: route_id, version, line (geography LineString), valid_from.
- route_bus_assignments: route_id, bus_id, trip_type (MORNING, AFTERNOON), weekday_mask, valid_from, valid_to.
- timetables: route_id, trip_type, departure_time (campus local time), weekday_mask, valid_from, valid_to.

10.4 People and assignments

- driver_roster: bus_id, driver_id, valid_from, valid_to.
- faculty_assignments: bus_id, faculty_id, is_primary, valid_from, valid_to.
- student_routes: student_id, route_id, trip_type, boarding_stop_id, valid_from, valid_to.
- student_profiles: user_id, department, year_of_study, date_of_birth (restricted column, encrypted at rest), guardian_contact (optional, encrypted at rest).

10.5 Operations: shifts, trips, samples, and state

- shifts: bus_id, driver_id, trip_id, phone_device_ref, mode_at_start, started_at, ended_at, end_reason (MANUAL, AUTO_IDLE, ADMIN, TIMEOUT), status (ACTIVE, ENDED).
- trips: route_id, bus_id, shift_id, trip_type, service_date, scheduled_departure, actual_departure, ended_at, status (SCHEDULED, ACTIVE, COMPLETED, CANCELLED).
- location_samples: source (HW, PHONE), bus_id, device_id (nullable), driver_id (nullable), shift_id (nullable), boot_id (nullable), seq, recorded_at, received_at, lat, lng, geom (geography), speed_kmh, heading_deg, accuracy_m, satellites, hdop, gps_valid, ignition, accepted (boolean), reject_reason. Partitioned by day. Unique on (device_id, boot_id, seq) and on (driver_id, shift_id, seq).
- bus_state: bus_id (primary key), state, active_source (HW, PHONE, NONE), mode, last_hw_valid_at, last_phone_valid_at, last_lat, last_lng, last_speed_kmh, last_heading_deg, consecutive_hw_valid, last_switch_at, version.
- source_switches: bus_id, from_source, to_source, reason_code, actor_type (SYSTEM, DRIVER, ADMIN), actor_id, at.
- device_events: device_id, bus_id (nullable), type, occurred_at, received_at, payload (JSON), status (RECORDED, PROCESSED, IGNORED), dedupe_key (unique, built as device_id, boot_id, event seq).
- stop_events: trip_id, stop_id, event (ARRIVED, DEPARTED), at, method (AUTO, MANUAL), source.

10.6 Attendance

- attendance_records: trip_id, student_id, status (EXPECTED, BOARDED, LATE_BOARDED, ABSENT, ON_LEAVE, MANUAL_CHECKIN, FLAGGED_GUEST), boarded_at, method (QR, DRIVER, RFID), marked_by, boarding_stop_id. Unique on (trip_id, student_id).
- leave_records: student_id, service_date, trip_type, reason_category, created_by, cancelled_at.
- absence_alerts: student_id, service_date, morning_trip_id, afternoon_trip_id, alerted_at, recipients (array), status.

10.7 Emergencies

- sos_incidents: bus_id, trip_id (nullable), first_trigger_type (DRIVER_APP, HW_BUTTON, STUDENT_APP, FACULTY_APP, SECURITY_APP), first_triggered_at, status (OPEN, RESPONDING, RESOLVED, CANCELLED, DRILL_CLOSED), current_tier, resolved_at, resolution_note, is_drill.
- sos_events: incident_id, type (TRIGGERED, JOINED, DELIVERED, ACKNOWLEDGED, ESCALATED, CANCELLED, RESOLVED, NOTE, DEVICE_RECEIVED), actor_id, at, detail (JSON). Append-only.

10.8 Stop requests

- stop_requests: requested_by, proposed_name, location (geography), route_id, status (PENDING, APPROVED, REJECTED, MERGED), merged_into_id, reviewed_by, review_note, approved_stop_id, effective_from.
- stop_request_votes: request_id, student_id. Unique on (request_id, student_id).

10.9 Notifications, announcements, and settings

- notifications: recipient_id, category, title, body, data (JSON), read_at.
- notification_deliveries: notification_id, channel (PUSH, SMS, EMAIL, IN_APP), status (QUEUED, SENT, DELIVERED, FAILED, SUPPRESSED), attempts, provider_ref, last_error.
- notification_preferences: user_id, category, channel, enabled.
- announcements: title, body, scope_type (ALL, ROUTE, BUS), scope_id, created_by, published_at, expires_at.
- settings: key (primary key), value (JSON), updated_by.
- feature_flags: key, enabled, scope, updated_by.
- sms_templates: purpose, dlt_template_id, body_with_placeholders, approved_at.

10.10 Compliance and audit

- audit_logs: actor_id, actor_role, action, entity_type, entity_id, before (JSON), after (JSON), ip, user_agent, at. Append-only.
- data_requests: user_id, type (EXPORT, CORRECTION, DELETION, WITHDRAW_CONSENT), status (RECEIVED, IN_PROGRESS, COMPLETED, REJECTED_WITH_REASON), received_at, due_at, completed_at.

- consents: user_id, purpose, notice_version, granted_by (self or guardian), granted_at, withdrawn_at.
- notice_versions: version, text_url, effective_from.

10.11 Indexes and constraints

- location_samples: index on (bus_id, recorded_at DESC). GIST index on geom.
- device_assignments: partial unique index on device_id where valid_to is null. Partial unique index on bus_id where valid_to is null.
- attendance_records: unique on (trip_id, student_id).
- device_events: unique on dedupe_key.
- sessions: index on (user_id, revoked_at).
- audit_logs: index on (entity_type, entity_id, at).
- Partitions: location_samples and device_heartbeats are partitioned by day. The partition manager creates partitions 7 days ahead. The retention job drops partitions older than the retention period (section 29).

11. Hardware and driver-phone integration contract

The API accepts data from three senders: hardware trackers, driver phones, and the device health channel. This section is the contract that the hardware and mobile teams build against. Changes to it follow section 11.9.

11.1 Device authentication

- Each device has a unique device_id in the form HW_ followed by 3 or 4 digits. A device has one active token at a time.
- The server generates a 256-bit random token when a technician provisions the device. The technician flashes it to the device over the serial port. The server stores only an HMAC-SHA256 hash of the token, keyed with a server secret.
- The token is sent only in the header Authorization: Bearer followed by the token. It is never sent in a URL or in a request body.
- Rotation with overlap: a coordinator or technician requests rotation. The server creates a PENDING token. The device calls POST /api/v1/devices/token/rotate with its current token and receives the new token once. The device stores it and uses it. The

first successful request with the new token activates it and revokes the old one. If the new token is not used within 24 hours, the PENDING token expires and the old token stays active.

- Revocation: after a coordinator revokes a token, the next request returns 401 DEVICE_REVOKED. The device stops sending and shows its error LED. The system opens a maintenance ticket automatically. A revoked device cannot receive a new token over the network, so restoring it needs physical reprovisioning.
- Unlinked device: a device with an active token but no bus link is accepted. Its data is stored with bus_id set to null and is not shown to users. The admin console lists unlinked devices as a warning.
- Inactive device (status REVOKED or RETIRED): every request returns 403 DEVICE_INACTIVE.

11.2 Endpoints

Endpoint	Sender	Auth	Purpose
POST /api/v1/telemetry	Hardware	Device token	Location samples and events
POST /api/v1/heartbeat	Hardware	Device token	Health report: every 60 s with ignition on, every 10 min when parked
POST /api/v1/devices/token/rotate	Hardware	Device token	Receive a rotated token
POST /api/v1/driver/locations	Driver phone	Driver session, shift-scoped	Phone location samples
POST /api/v1/driver/status	Driver phone	Driver session	Phone health and permissions

11.3 Telemetry request

Example request:

```
{
  "schema_version": 1,
  "device_id": "HW_001",
  "boot_id": "a3f19c02",
```

```
  "fw_version": "1.4.2",
  "sent_at": "2026-10-08T10:32:16Z",
  "samples": [
    {
      "seq": 10451,
      "recorded_at": "2026-10-08T10:32:15Z",
      "gps_valid": true,
      "lat": 12.345678,
      "lng": 76.543210,
      "speed_kmh": 32.4,
      "heading_deg": 182.0,
      "satellites": 8,
      "hdop": 1.2,
      "ignition": "ON",
      "vehicle_battery_mv": 13620
    }
  ],
  "events": []
}
```

Field rules:

Field	Type	Required	Rules
schema_version	integer	Yes	Must be 1 for this version
device_id	string	Yes	Pattern HW_ followed by 3 or 4 digits. Must match the token
boot_id	string	Yes	8 to 16 lowercase hex characters. New value on every boot
fw_version	string	Yes	Semantic version, for example 1.4.2
sent_at	string	Yes	UTC, ISO 8601. Used only for clock-skew detection
samples	array	Yes (may be empty)	0 to 50 items

Field	Type	Required	Rules
samples[].seq	integer	Yes	0 or more. Increases by 1 per sample within a boot_id. Gaps are allowed
samples[].recorded_at	string or null	Yes	UTC from GNSS when gps_valid is true, otherwise null
samples[].gps_valid	boolean	Yes	True only when the fix meets the device's own quality check
samples[].lat	number or null	Yes	-90 to 90, at most 6 decimal places. Null when gps_valid is false
samples[].lng	number or null	Yes	-180 to 180, at most 6 decimal places. Null when gps_valid is false
samples[].speed_kmh	number or null	Yes	0 to 250
samples[].heading_deg	number or null	No	0 to 360
samples[].satellites	integer or null	Yes	0 to 40
samples[].hdop	number or null	Yes	0.5 to 99
samples[].ignition	string	Yes	ON, OFF, or UNKNOWN
samples[].vehicle_battery_mv	integer or null	No	0 to 60000
events	array	Yes (may be empty)	0 to 20 items. Shape in section 12

The request body is limited to 8 KB. Larger bodies return 413.

11.4 Responses

Successful response example:

```
{
  "schema_version": 1,
  "ack_seq": 10451,
  "rejected": [{"seq": 10449, "reason": "STALE_TIMESTAMP"}],
  "event_results": [{"seq": 7, "status": "RECORDED", "incident_id": null}],
  "config": {"moving_interval_s": 10, "stopped_interval_s": 30,
"parked_heartbeat_s": 600},
  "server_time": "2026-10-08T10:32:16Z"
}
```

Ack rule: ack_seq is the highest sequence number for which every earlier sample in the batch was either accepted or rejected with a reason. The device deletes buffered samples up to and including ack_seq. Rejected samples are not resent.

HTTP status	Meaning	Device action
200	Processed. Body contains ack_seq, rejected, event_results, config, server_time	Delete acknowledged samples. Apply config
400	Body does not match the schema. Body lists field errors. Nothing is stored	Do not resend the same body. Move the batch to the device error log and continue
401	Token missing, wrong, or revoked (DEVICE_REVOKED)	Stop uploading. Show the error LED. Check again every 15 minutes
403	Device inactive, or device_id does not match the token	Same as 401
413	Body larger than 8 KB	Split the batch in half and resend each half
429	Rate limited. Retry-After header is present	Wait for Retry-After seconds. Keep buffering

HTTP status	Meaning	Device action
500, 502, 503, 504	Server error or maintenance	Retry with backoff of 5, 10, 20, 40, 80, then 120 seconds, with plus or minus 20 percent jitter. Honour Retry-After if present
No response (timeout)	Outcome unknown	Resend the same body. The server deduplicates on device_id, boot_id, and seq

11.5 Validation pipeline

The server applies these checks in order:

1. Authenticate the token. On failure return 401 or 403.
2. Confirm the device status is ACTIVE.
3. Check the body size (8 KB limit). On failure return 413.
4. Validate the schema. If the body is invalid, return 400 with field errors and store nothing.
5. Apply the range rules in section 11.3 to each sample. An out-of-range sample is rejected alone, with reason OUT_OF_RANGE.
6. Check timestamps. A recorded_at more than 60 seconds in the future is rejected with FUTURE_TIMESTAMP. A recorded_at older than 7 days is rejected with STALE_TIMESTAMP. A recorded_at older than 10 minutes is accepted and marked BUFFERED. Buffered samples update history only, never live state.
7. Check for duplicates. If (device_id, boot_id, seq) exists with identical content, accept silently. If it exists with different content, reject with CONFLICT.
8. Reject null island: lat equal to 0 and lng equal to 0 is rejected with NULL_ISLAND.
9. Check the service area (default: campus and route corridors plus 10 km). Outside the area, store the sample flagged and reject it with OUT_OF_AREA.
10. Check plausibility. If the distance from the last accepted valid sample, divided by the elapsed time, exceeds 150 km/h, reject with IMPLAUSIBLE_JUMP and store it. If more than 30 minutes have passed since the last accepted sample, reset the reference point and skip this check for that sample.
11. Check clock skew. If the absolute difference between sent_at and server time exceeds 300 seconds, record CLOCK_SKEW for the device and raise a technician alert. Ordering still uses received_at.

12. Store all accepted samples, all rejected samples with their reasons (duplicates excepted), and all events in one database transaction.
13. Queue a source-selection evaluation for the bus linked to the device, if there is one.
14. Build the response.

11.6 Heartbeat

Field	Type	Rules
device_id, boot_id, fw_version, sent_at	As in 11.3	Same rules
uptime_s	integer	Seconds since boot
network_status	string	REGISTERED, SEARCHING, DENIED, NO_SERVICE, or UNKNOWN
sim_status	string	READY, NOT_INSERTED, PIN_REQUIRED, LOCKED, or UNKNOWN
signal_rssi_dbm	integer	-125 to 0
vehicle_battery_mv	integer or null	0 to 60000
device_battery_mv	integer or null	0 to 10000
gps_status	string	FIX_VALID, NO_FIX, MODULE_ERROR, or DISABLED
buffer_bytes, buffer_capacity_bytes	integer	Current use and capacity of the offline buffer
ignition	string	ON, OFF, or UNKNOWN
temperature_c	number or null	-40 to 85. Optional

Response: schema_version, server_time, config, and pending_action. pending_action is one of NONE, UPDATE_CONFIG, REBOOT, or FIRMWARE_UPDATE. When it is FIRMWARE_UPDATE, the response also includes firmware_target with the version to install.

Device health, computed on the server: ONLINE means the last heartbeat is within twice the expected interval. DELAYED means it is older than that but within 10 minutes while ignition is on, or within 24 hours while parked. OFFLINE means it is older than 10 minutes while ignition is on or during a shift, or older than 24 hours while parked.

11.7 Driver-phone endpoints

POST /api/v1/driver/locations

- Auth: driver session token that carries a shift_id. The shift must be ACTIVE, and the driver must be on the roster of that shift's bus (R-16).
- Body fields: shift_id, app_version, device_model (up to 64 characters), samples (1 to 50 items). Each sample has seq (increasing per shift), recorded_at (UTC from the phone), lat, lng, accuracy_m (0 to 1000), speed_mps (nullable), heading_deg (nullable), provider (GPS, FUSED, NETWORK, or UNKNOWN), and is_mock (boolean).
- A phone sample updates live state only if accuracy_m is 50 or less (default), provider is not NETWORK, is_mock is false, and it passes the checks in 11.5. Mock locations are stored and flagged, and they never update live state.
- speed_mps is converted to km/h (multiplied by 3.6) on the server.
- The body is limited to 16 KB.
- Response: the same ack structure as telemetry, plus phone_interval_s (30 when the active source is HW, 10 when it is PHONE) and active_source for the bus.
- Android reports mock locations directly. iOS detection is less reliable, so is_mock is best effort on iOS.

POST /api/v1/driver/status

- Body fields: app_version, os (ANDROID or IOS), os_version, location_permission (ALWAYS, WHILE_IN_USE, or DENIED), notifications_permission (GRANTED or DENIED), battery_percent, battery_optimisation_exempt (Android only; null on iOS), background_running (boolean), gps_enabled (boolean), and network (WIFI, CELLULAR, or NONE).
- Used for the pre-trip check (section 15), driver warnings, and phone health in the admin console.
- Sent at shift start, every 5 minutes during a shift, and whenever a permission changes.

11.8 Rate limits

Sender	Sustained limit	Burst	When exceeded
Device telemetry	1 request per 2 seconds	5	429 with Retry-After
Device heartbeat	1 request per 30 seconds	3	429 with Retry-After
Driver locations	1 request per 5 seconds	10	429 with Retry-After
Driver status	1 request per 60 seconds	3	429 with Retry-After
All ingestion combined	200 requests per second (default)	Not applicable	503 with Retry-After

11.9 Versioning and compatibility

- The path /api/v1 stays supported while any device reports schema_version 1.
- Within one version, changes are additive only: new optional request fields and new response fields. The server ignores unknown request fields. Devices ignore unknown response fields.
- A breaking change requires /api/v2. Both versions run side by side. Retiring a version requires 90 days' notice and zero devices reporting that version. The admin console shows device counts per schema version.
- Contract tests replay recorded messages from every released firmware version on every backend build (section 32).

11.10 Ordering and time

- Sequence checks use (device_id, boot_id, seq). Arrival order uses received_at.
- Live state is the accepted valid sample with the latest recorded_at. Ties go to the higher seq. Older samples that arrive late update history only.
- Samples from an earlier boot_id are accepted and ordered by recorded_at.
- Server time is authoritative for every age calculation shown to users.

12. Hardware and phone events

Events are separate from samples. Each event has type, seq (per boot), occurred_at (UTC device time, or null if unknown), and data. Events are stored before their actions run. SOS is processed before any other event in the same request. An event with a duplicate dedupe key is stored once. Events older than 7 days are stored but not acted on.

Event	Sent when	Data fields	What the server does	Who is alerted	Device behaviour
BOOT	After every power-up or reset	reset_reason (POWER, WATCHDOG, BROWNOUT, FIRMWARE, UNKNOWN), fw_version	Logged. Three WATCHDOG or BROWNOUT events in 24 hours open a maintenance ticket	Technician (ticket)	Resume normal operation
IGNITION_ON	Ignition input changes to ON	None	Ends parked state for the bus. Starts the shift timer if a shift is active	None	Switch to moving or stopped cadence
IGNITION_OFF	Ignition input changes to OFF	None	If a shift is active, starts the idle timer for auto-end (section 15)	None	Switch to parked cadence
SOS	SOS button held for 300 ms (debounced)	triggered_at (UTC), last_fix (lat, lng, recorded_at, gps_valid)	Creates an incident, or joins an open incident for the same bus within 60 seconds (section 19). Starts tier 0 fan-out immediately	Tier 0 recipients immediately, then per section 19	Resend every 5 seconds until a response returns incident_id. Keep the LED solid and the buzzer on until then
GPS_LOST	No valid fix for 30 seconds while ignition is ON	last_valid_at	Marks the hardware fix invalid for the selector. Logged	Ticket if it lasts more than 10 minutes	Keep trying to acquire a fix
GPS_RECOVERED	First valid fix after GPS_LOST	outage_s	Logged. Counts toward the 3 consecutive valid fixes needed to return to HW	None	Resume normal cadence

Event	Sent when	Data fields	What the server does	Who is alerted	Device behaviour
NETWORK_RESTORED	Cellular registration returns after a loss	outage_s, buffered_samples	Logged as a network outage with its duration	None	Flush the buffer, oldest first
BUFFER_OVERFLOW	Buffer full and oldest samples dropped	dropped_count, from_seq, to_seq	Logged. Repeated overflows open a ticket	Technician (ticket)	Continue
LOW_VEHICLE_BATTERY	vehicle_battery_mv below the class threshold for 30 seconds. Defaults: 11800 mV (12 V class), 23500 mV (24 V class)	mv, vehicle_class	Opens a ticket	Technician and coordinator	None
LOW_DEVICE_BATTERY	Backup battery below threshold	mv	Opens a ticket	Technician (ticket)	None
MODULE_ERROR	Cellular module reports an error or resets unexpectedly	error_code	Logged. Three in 24 hours open a ticket	Technician (ticket)	Attempt module recovery
CONFIG_APPLIED	Device applies configuration from a response	config_version	Logged	None	None
FIRMWARE_UPDATED	New firmware boots after an update	from_version, to_version, ok	Updates the device firmware record. Confirms the rollout step (section 33)	None	None
FIRMWARE_FAILED	Update failed and the device rolled back	attempted_version, reason	Opens a ticket. Pauses the rollout for this device	Technician	Keep the previous firmware
TAMPER	Tamper switch triggered (only if fitted, see HW-SPEC)	occurred_at	Opens a ticket and raises an alert	Coordinator and technician	None
DOOR_OPEN, DOOR_CLOSE	Door sensor changes (only if fitted)	None	Stored. Available to stop confirmation in phase 2	None	None

Event	Sent when	Data fields	What the server does	Who is alerted	Device behaviour
RFID_TAP	Card tapped on the RFID reader (phase 2)	card_uid, hashed on receipt	Used by attendance when the RFID method is enabled (section 18)	None	None
CLOCK_SYNC	Device clock corrected	old_time, new_time	Logged	None	None
Unrecognised type	Any other type	Not applicable	Stored with status IGNORED and returned in event_results	None	Continue

Server-generated events, which hardware never sends: NO_HEARTBEAT, NO_SAMPLES, SOURCE_SWITCHED, SHIFT_STARTED, SHIFT_ENDED, TRIP_LATE_START, and SOS_ESCALATED. They appear in the operations timeline and the audit log.

13. Source selection

The server decides which source (HW or PHONE) is active for each bus. Clients never decide. Exactly one source is active at a time. The two sources are never averaged or blended. Both are stored in full, so the admin console can compare them.

13.1 Valid fix definitions

- Hardware fix is valid when gps_valid is true, satellites is at least 4 (default), hdop is 5.0 or less (default), the sample passed section 11.5, and it is not BUFFERED.
- Phone fix is valid when accuracy_m is 50 or less (default), provider is not NETWORK, is_mock is false, and the sample passed section 11.5.
- A source is fresh when its last valid fix is no older than 60 seconds (default).

13.2 Modes

- AUTO (default): the server switches sources as described below.
- MANUAL_HW: the hardware is used. The server does not switch to the phone automatically. If the hardware is not fresh, the bus shows as STALE and then UNAVAILABLE.
- MANUAL_PHONE: the phone is used. The server does not switch back automatically.

- The driver sets the mode in the app. The coordinator can override it from the console. Every change is logged with the actor and reason.
- A new shift starts in AUTO unless the coordinator sets a different default for that bus.

13.3 Evaluation

The selector runs on every accepted sample for a bus, and every 5 seconds for each bus with an active shift.

```
RUN FOR BUS B:
1  If B has no ACTIVE shift: state = NOT_RUNNING, active = NONE. Stop.
2  hw_fresh    = last_hw_valid_at is within 60 s of now
3  phone_fresh = last_phone_valid_at is within 60 s of now
4  cooldown_ok = last automatic switch was more than 30 s ago
5  Apply the rule for the current mode:
   MANUAL_HW:    active = HW.   If not hw_fresh, state = STALE.
   MANUAL_PHONE: active = PHONE. If not phone_fresh, state = STALE.
   AUTO:
     active = HW and not hw_fresh and phone_fresh and cooldown_ok  -> switch
to PHONE, reason HW_STALE
     active = PHONE and consecutive_hw_valid >= 3 and cooldown_ok  -> switch
to HW, reason HW_RECOVERED
     active = PHONE and not phone_fresh and hw_fresh and cooldown_ok ->
switch to HW, reason PHONE_LOST
     active = NONE and hw_fresh                                    -> switch
to HW, reason INITIAL
     active = NONE and phone_fresh                                 -> switch
to PHONE, reason PHONE_FALLBACK
6  If neither source is fresh: state = STALE.
   If neither source has been fresh for 120 s: state = UNAVAILABLE, and raise
one alert.
7  Otherwise: state = LIVE, with the active source.
```

consecutive_hw_valid counts valid hardware fixes in a row. It resets on any invalid sample. Three valid fixes within 30 seconds are required before the source returns to HW.

13.4 Transitions

From	To	Trigger	Reason code	Driver sees	Coordinator sees
NONE	HW	First valid hardware fix after shift start	INITIAL	Live, bus tracker	Logged
NONE	PHONE	Hardware not fresh, phone fresh	PHONE_FALLBACK	Banner: using your phone's GPS	Alert
HW	PHONE	Hardware stale for more than 60 s, phone fresh, cooldown passed	HW_STALE	Banner	Alert within 30 seconds
PHONE	HW	Three valid hardware fixes within 30 s, cooldown passed	HW_RECOVERED	Banner: bus tracker is back	Logged
PHONE	HW	Phone not fresh, hardware fresh	PHONE_LOST	Banner	Logged
Any	MANUAL_HW	Mode change by driver or coordinator	DRIVER_MODE or ADMIN_OVERRIDE	Confirmation	Logged with actor
Any	MANUAL_PHONE	Mode change by driver or coordinator	DRIVER_MODE or ADMIN_OVERRIDE	Confirmation	Logged with actor
Any	NONE	Shift ends	SHIFT_ENDED	Shift summary	Logged

Manual mode changes bypass the cooldown. A manual change to a source that is not fresh leaves the bus in state STALE until that source recovers.

13.5 Anti-flapping rules

- Automatic switches are at least 30 seconds apart (cooldown).
- Returning to HW needs three valid fixes within 30 seconds.
- Every switch is written to source_switches with the reason code.
- If a bus switches more than 6 times in 10 minutes, the server stays on the current source for 5 minutes and raises a technician alert.

13.6 Phone cadence

- Standby: while HW is active, the phone sends every 30 seconds.
- Fallback: while PHONE is active, the phone sends every 10 seconds.
- The server states the required interval in every driver-phone response (phone_interval_s). The phone follows that value.

13.7 Labels shown to users

Bus state	Label
LIVE, active HW	Live, bus tracker
LIVE, active PHONE	Live, driver's phone
STALE	Last update N seconds ago. Shown grey
UNAVAILABLE	Location unavailable. Last seen at HH:MM
NOT_RUNNING	Not running. Next trip at HH:MM, when scheduled

The source label is always shown beside the position, never only in a menu. Colour is never the only way a state is shown: each state also has text and an icon.

13.8 Comparison view

The admin console shows the last 10 minutes of both sources as two tracks on one map, with the active source marked. This view is for diagnosis and is available to COR and SAD only.

14. Bus state model

14.1 States

State	Condition	Student and faculty text	Admin text
NOT_RUNNING	No ACTIVE shift for the bus	Not running. Next trip at HH:MM	Not running (device health shown separately)

State	Condition	Student and faculty text	Admin text
LIVE	ACTIVE shift and active source fresh	Live label from 13.7	Source, age in seconds, mode
STALE	ACTIVE shift and active source not fresh, but not yet 120 s without any fresh source	Last update N seconds ago	Stale, with reason
UNAVAILABLE	ACTIVE shift and no fresh source for 120 seconds	Location unavailable. Last seen at HH:MM	Alert open
NO_HARDWARE	ACTIVE shift, no device linked to the bus, mode AUTO	Live, driver's phone	Notice: no hardware linked

NOT_RUNNING is never shown as Offline (R-20). Device health is shown separately (section 11.6).

14.2 Device health states

Health	Meaning	Where shown
ONLINE	Heartbeat within twice the expected interval	Technician console, admin bus card
DELAYED	Heartbeat late but within the offline threshold	Technician console
OFFLINE	Heartbeat past the offline threshold (section 11.6)	Technician console, alert

15. Shifts and trips

15.1 Pre-trip check (driver app)

#	Check	Source	Blocks start?	Rule
1	Driver signed in and rostered for the chosen bus	Server	Yes	Only rostered buses are listed
2	Trip type selected (MORNING or AFTERNOON)	Driver	Yes	Pre-selected from the timetable when a trip is due
3	Hardware status	Server	No, warning only	GREEN: heartbeat within 2 minutes and gps_status FIX_VALID. AMBER: heartbeat within 2 minutes, no fix. RED: no heartbeat within 2 minutes. Start is allowed with a warning and the phone becomes primary
4	Phone location permission set to Always (Android background permission; iOS Always)	Phone	Yes	Fix screen shows the steps for the phone's OS
5	Notification permission granted	Phone	Yes	SOS depends on notifications
6	Battery optimisation exempt (Android only)	Phone	No, warning	Opens the system setting directly
7	Phone GPS fix with accuracy 50 m or better within 60 seconds	Phone	Yes, unless hardware is GREEN	Waits and shows accuracy while waiting
8	Network reachable	Phone	Yes	One successful driver status call

#	Check	Source	Blocks start?	Rule
9	Confirm start	Driver	Not applicable	Start button enabled only when all blocking checks pass

15.2 Start effects

- Create the shift with status ACTIVE. Create the trip for this bus, route, service date, and trip type, or reuse the existing one.
- Block the start if another shift is ACTIVE for the same bus, and show who holds it.
- Set the bus mode to AUTO, or to the bus default if one is set.
- Create one EXPECTED attendance record for each student with an active route assignment for that route and trip type, unless the student is ON_LEAVE.
- Start phone standby sampling. The driver app runs its background task (section 16.3).
- Notify faculty assigned to the bus: trip started (category TRIP).

15.3 Running rules

- A driver can run only one shift at a time (R-03).
- If the driver app stops during a shift, the server marks the phone source as not fresh and the selector takes over. When the app restarts, it resumes the same shift_id automatically. It does not create a new shift.
- The driver's session expires 12 hours after sign-in. A new sign-in during an active shift resumes that shift.

15.4 End rules

End reason	Trigger	Behaviour
MANUAL	Driver taps End and confirms	Immediate
AUTO_IDLE	No movement below 3 km/h for 20 minutes and ignition OFF (or no hardware and no phone movement for 20 minutes)	Driver is prompted. If there is no response within 10 minutes, the shift ends with AUTO_IDLE
TIMEOUT	Shift older than 14 hours	Ends with TIMEOUT and raises a coordinator alert

End reason	Trigger	Behaviour
ADMIN	Coordinator ends the shift with a reason	Immediate, logged with actor

Rules that apply to every end:

- The driver cannot end a shift while an SOS incident for that bus is OPEN. Only the coordinator can end it in that case, with a reason.
- Ending a shift stops phone tracking, marks the trip COMPLETED, computes the trip summary, sets the bus state to NOT_RUNNING, and closes the boarding window.
- QR codes for the trip stop being generated once the trip ends.

15.5 Trip generation

- A nightly job creates trip records for the next 7 days (default) from route_bus_assignments and timetables.
- A trip is started from its generated schedule. An unscheduled trip is allowed only when the coordinator enables that setting.
- The coordinator can cancel a trip. Cancelling notifies faculty and students on the route (category TRIP).

15.6 Late start

- If no shift starts for a bus within 10 minutes of its scheduled departure, the server records TRIP_LATE_START and notifies faculty assigned to the bus.
- If the trip is still not started 20 minutes after departure, the coordinator is notified too.

15.7 Trip summary

Computed when the trip is completed:

- Distance covered from accepted valid samples, in kilometres to one decimal place.
- Duration from first to last valid sample.
- Average moving speed, excluding samples below 3 km/h.
- Stops arrived at and departed from, with times.
- Boardings: count by method (QR, DRIVER, RFID), and late boardings.
- Source usage: percentage of trip time on HW, PHONE, and no active source.
- Failovers: count of automatic source switches.

- Longest data gap, in minutes.
- SOS incidents linked to the trip.

16. Features by role

Each role lists its screens, its workflows (numbered), its rules, and its acceptance criteria (AC). Workflows reference the cross-cutting sections (13 to 15, 17 to 23) rather than repeat them.

16.1 Student (STU)

Screens: sign-in, home (buses on my route), bus detail, my route and stops, boarding (QR scanner), attendance history, leave, stop request, notifications, profile and privacy, help, SOS.

Workflows:

1. First sign-in: college SSO, then match to the imported record. Confirm name and department. The route is shown as set by the coordinator. If no route is set, the screen says: your route is not set, contact the transport office.
2. Notification permission is requested with a short reason. The app does not request location permission.
3. Consent: the privacy notice is shown and accepted. Students under 18 see the parental consent flow (section 29) before any feature is enabled.
4. Track a bus: the home screen lists buses serving my route with an active or scheduled trip today. Tapping one opens the bus detail: source label, data age, speed, next stop, ETA to my stop, and remaining stops.
5. Approaching alert: sent once per trip when the ETA to my stop is at or below my threshold. Default 5 minutes, adjustable from 2 to 15 minutes. Not sent if I am on leave or already boarded.
6. Board the bus: at the bus, tap Board bus. The camera scans the QR code shown by the driver. On success the screen shows: Boarded Bus 01 at 07:42.
7. Boarding failures show a specific message: QR expired (scan the current code), This bus is not on your route today, You are already boarded, Boarding window closed (ask the driver to check you in), or No connection (ask the driver to check you in).
8. Attendance history: a list by date, showing morning and afternoon status for each trip.
9. Mark leave: choose one or more dates and trip types. Allowed until 30 minutes before the trip's scheduled departure (default). After that, contact faculty.

10. Request a new stop: section 20.
11. SOS: section 19. Hold for 2 seconds to send.
12. Notification preferences: section 21.
13. Profile and privacy: view name, department, and route; request a correction; export my data; request deletion; view consent history; list active sessions; sign out.
14. Offline behaviour: the last known bus positions stay visible for up to 10 minutes, marked as offline. Boarding needs a connection and is never queued. SOS retries every 10 seconds. If the phone has no data connection, the screen shows the security office number (set in configuration).

Rules:

- A student sees only buses serving their route today (R-12).
- A student's own location is never sent to or shown by the server (R-19).
- A student cannot see another student's record, attendance, or leave.

Acceptance criteria:

- AC-STU-01: Given an active trip on my route, when I open the home screen, the bus appears within 3 seconds at the 95th percentile, with its source label.
- AC-STU-02: Given a bus that does not serve my route, when I request its detail, the response is 404 and the bus is not listed.
- AC-STU-03: Given a valid QR code scanned inside the boarding window, a single boarding record is created. A second scan returns DUPLICATE_BOARDING with the original time.
- AC-STU-04: Given the boarding window has closed, a scan returns BOARDING_WINDOW_CLOSED. A driver check-in with reason LATE records LATE_BOARDED.
- AC-STU-05: A student's API responses never contain location data for any other person.

16.2 Faculty (FAC)

Screens: my buses, bus detail (live map, driver name and call button, source label, trip progress), today's attendance (morning and afternoon, per student), alerts inbox, SOS console, leave list, reports for my buses, settings, help.

Workflows:

1. See the buses I am assigned to. The primary faculty member for each bus is marked.

2. Monitor a trip: live position, next stop, ETA to every remaining stop in order, and boardings so far against expected passengers.
3. Respond to an absence alert (rule in section 18.6): receive the push and in-app alert, open the student's attendance, add a note, and mark the alert acknowledged.
4. Respond to SOS: acknowledge, add notes, and resolve with a resolution note (section 19).
5. Mark leave for a student on my bus. The reason is a category only: Personal, Academic, or Other. No free-text reason is stored, so no health information enters the system through leave.
6. Export attendance for my buses as CSV for a date range of up to 31 days.
7. Call the driver from the bus detail screen (opens the phone dialler).
8. Receive trip notices: trip started, late start, delay, cancellation, and route change (categories TRIP and ADMIN).

Rules:

- Faculty see only buses they are assigned to (A scope). A request for another bus returns 404.
- A faculty member assigned to several buses sees a bus picker at the top of every screen.
- Attendance exports are logged in the audit log with row counts.

Acceptance criteria:

- AC-FAC-01: Given I am not assigned to Bus 05, a request for Bus 05 returns 404.
- AC-FAC-02: Given I am assigned to two buses, both appear, and the bus picker switches the view without a reload.
- AC-FAC-03: Given an absence condition is met at the cutoff, the alert reaches my app within 5 minutes of the cutoff.

16.3 Driver (DRV)

Screens: sign-in (OTP), roster (buses I can run), pre-trip check, live shift (map, next stop, mode, source health, boarding QR, check-in list, SOS, end shift), trip summary, report a problem, settings (permissions, battery status, consent notice), help.

Workflows:

1. Sign in with the email OTP (section 8). The app registers this phone as the driver's phone (device label recorded). Location and notification permissions are requested on this screen, with the reason shown first.
2. Run the pre-trip check (section 15.1).
3. Start the shift (section 15.2).
4. Drive: the map shows my position, the route line, the next stop, and the ETA. Stop arrivals are detected automatically (section 17.4). If detection fails, the driver can tap Mark arrived.
5. Boarding check-in: the screen shows the boarding QR code, which rotates every 30 seconds and shows a countdown. The check-in list lets the driver record a boarding manually. A manual check-in needs a reason when the student is not on the expected list.
6. Mode switch: a control with three options (Auto, Hardware, Phone). Choosing one opens a confirmation sheet. An optional note of up to 140 characters is stored with the change.
7. SOS: hold for 2 seconds, choose a category (Emergency, Medical, Breakdown, Other), add an optional message, and send. The screen then shows the incident status: received, faculty notified, acknowledged.
8. Report a problem: categories are Bus breakdown, Tracker problem, Phone problem, and Other. Free text is limited to 500 characters. Creates a maintenance ticket (section 16.5).
9. End the shift (section 15.4).
10. Offline: the phone buffers location for up to 24 hours (about 8,640 samples at 10-second intervals). A banner shows Sending saved location, N points. Buffered samples follow the rules in section 11.5.
11. Tracking indicator: while a shift runs, the app shows a persistent indicator on screen and a persistent notification (the Android foreground service notification, and the iOS status indicator).
12. Settings: permission status, battery optimisation status, phone model, app version, and sign out. Sign-out is blocked during an active shift.

Rules:

- A driver sees only buses on their roster. A request for another bus returns 403.
- Phone location is collected only while a shift is active (R-18).
- The driver app never shows other drivers' rosters or positions.

Acceptance criteria:

- AC-DRV-01: Given I am not rostered on Bus 07, Bus 07 is not listed, and a direct request returns 403.
- AC-DRV-02: Given the hardware status is RED, the shift can start with a visible warning and the phone becomes the primary source.
- AC-DRV-03: Given the phone does not have Always location permission, the start is blocked, and the fix screen shows the steps for this phone.
- AC-DRV-04: Given the app is closed mid-shift and reopened, the same shift resumes within 10 seconds, with no duplicate shift.

16.4 Coordinator (COR), web console

Pages: dashboard, live fleet map, buses, devices, routes and stops, timetables, assignments (roster, faculty, student routes), users, stop requests, SOS console, alerts, source log, attendance reports, announcements, reports, audit log (read-only), settings (read-only, except thresholds for SAD).

Workflows:

1. Create a bus: bus number, registration number, capacity, and document expiry dates for insurance, fitness, pollution, and permit. Reminders are sent to the coordinator 30 and 7 days before each expiry.
2. Link a device to a bus: choose a device that the technician has provisioned, and set the valid-from date. The previous link, if any, is closed on the same date.
3. Create a route: name, code, ordered stops from the stop library, route geometry (drawn on the map, or uploaded as GeoJSON), and bus assignments for MORNING and AFTERNOON by weekday.
4. Set timetables per route and trip type.
5. Roster drivers: invite by email, then assign to buses. A driver can be rostered on several buses.
6. Assign faculty to buses. One faculty member is marked primary.
7. Import students by CSV (section 36). The system shows a validation report before commit: rows that pass, and rows with errors listed by row number.
8. Review stop requests (section 20).
9. Handle SOS: view the incident timeline, add notes, assign a responder, resolve with a note, and export an incident report (section 19).
10. Override a bus mode, with a required reason.

11. Revoke a device token: type the bus number to confirm. The device status becomes REVOKED and a maintenance ticket is opened (section 11.1).
12. Deactivate a user, for example when a student graduates or a driver leaves. Enter an end date. Assignments close on that date, and sessions are revoked.
13. Publish an announcement: scope is all users, one route, or one bus. Fields are title, body (up to 1,000 characters), and expiry. Sends a push notification to the scope (category ADMIN).
14. Cancel a trip, with a reason. Notifies faculty and students on the route (category TRIP).
15. Download reports as CSV (section 23).
16. Review the source-switch log for a bus or date range (section 13).

Rules:

- Destructive actions (revoke, deactivate, delete, cancel trip) need a confirmation step and a reason.
- Every change is written to the audit log with the old and new values.
- Bulk actions are limited to 500 rows per request.
- The console is responsive down to tablet width. It is not designed for phones.

Acceptance criteria:

- AC-COR-01: Given a device is revoked, its next request is rejected within 5 seconds, and a ticket appears on the technician's list.
- AC-COR-02: Given a student CSV with 3 bad rows out of 400, the commit is blocked until the bad rows are fixed or removed, and the report lists the 3 row numbers.
- AC-COR-03: Given a deactivated user, the user disappears from all live views within 60 seconds.

16.5 Technician (TEC), web console

Pages: device health list, device detail (heartbeats, events, firmware, token status), provisioning, token rotation, firmware rollout, maintenance tickets, replacement wizard, bench test.

Workflows:

1. Provision a batch: enter or import device IDs with serial number, IMEI, and ICCID. The system creates PROVISIONED records. The device token is generated, shown once as a text code and a QR code for flashing, and stored only as a hash. A label is printed with the device ID and a provisioning QR code.

2. Verify an installation: after installation, the technician checks that a heartbeat arrives within 2 minutes, a valid GNSS fix appears within 5 minutes (outdoors), and telemetry samples arrive within 1 minute. The commissioning checklist (HW-SPEC section 15) is recorded in the console. The device shows as Ready to link only after sign-off.
3. Replace a device: a wizard asks for the bus, the old device, the reason (failed, lost, upgraded), and the new device. The old device becomes RETIRED or REVOKED. The new device is linked from the same timestamp. Samples from the old device stay attached to it.
4. Investigate a delayed or offline device: the list shows last heartbeat, last valid fix, vehicle and device battery, signal strength, SIM status, buffer use, and the last ten events.
5. Roll out firmware: upload a release (version, SHA-256, size). Rollout runs in steps: a canary group of 5 buses, then 25 percent of the fleet, then the full fleet. Rollout pauses automatically if more than 2 percent of updates return FIRMWARE_FAILED, or if any updated device goes OFFLINE within 30 minutes of its update (section 33).
6. Work tickets: assign, add notes, and close with a resolution note. A ticket open for more than 48 hours escalates to the coordinator.
7. Bench test: a device registered as BENCH is excluded from fleet views, alerts, and reports.

Rules:

- The technician cannot see passenger, attendance, or student data.
- The technician cannot link a device to a bus. That is a coordinator action (R-01, R-02).
- Token rotation can be requested for a device but not viewed. Token values are never displayed after provisioning.

Acceptance criteria:

- AC-TEC-01: Given a new device is provisioned, its token works once it is flashed, and the first heartbeat is recorded within 2 minutes of power-up.
- AC-TEC-02: Given a replacement, the old device's history remains visible, and no sample is ever attached to the wrong device.

16.6 Super admin (SAD)

All coordinator functions, plus:

- Manage roles and role assignment.
- Configure college SSO (domain, OpenID Connect client details) and the MFA policy.

- Edit thresholds (section 35) and feature flags. Each edit requires a reason and is audited.
- Manage notification templates and their DLT registration IDs.
- View backup and restore status (section 34).
- Read the audit log.
- Use the emergency access procedure: a break-glass account with MFA, used only during a sign-in outage, with every action logged. The procedure is tested every quarter.

Rules:

- At least two SAD accounts must exist. The system blocks removing a SAD account when only one other remains.
- A SAD cannot disable MFA for their own account.

16.7 Security staff (SEC), mobile and web

- Live map of all active buses, read-only, with source labels and state (section 14).
- Receive SOS alerts with the incident timeline, acknowledge, mark Responding, and resolve with a note (section 19).
- Incident history for any bus and date.
- No editing of buses, routes, users, or attendance.
- Every view and action is logged.

16.8 Management (MGT), web

- Dashboards with fleet on-time performance (trip start against schedule), trips completed, SOS count by week, device uptime summary, and attendance by route (counts and percentages only).
- Reports are aggregate only. Management cannot see individual location history, individual attendance records, or individual SOS details beyond date, bus, and resolution status.

17. Live tracking, ETA, and stop arrivals

17.1 Map

- Map library: MapLibre with OpenStreetMap tiles. The layer style is a configuration value so that a Google Maps layer can be added later.
- Bus marker: an icon with the bus number, coloured by route, with a direction arrow from the heading. The source badge sits beside it. A stale bus is drawn grey with its age.
- Clustering: markers within 40 pixels of each other are grouped. Tapping a cluster shows its buses in a list.
- Route line and stop markers come from the current route geometry version.

17.2 Update cadence

- Foreground apps request the snapshot every 10 seconds.
- Background apps do not poll in the MVP.
- On resume, the app requests the snapshot immediately.
- The web console polls every 10 seconds while the tab is visible.

17.3 Snapshot endpoint

GET /api/v1/snapshot?route_id=R1 returns, for each bus on the route with a trip today: bus_number, state, active_source, mode, lat, lng, heading_deg, speed_kmh, data_age_s, next_stop_id, eta_to_my_stop_s (for students, using the student's boarding stop), remaining_stops (ordered, with ETA), and trip_status. Responses are cached for 3 seconds per route per instance and carry an ETag, so an unchanged response returns 304.

The caller's route is taken from the session. A student or faculty member cannot request another route by changing the parameter. The API returns 404 for routes the caller cannot see.

17.4 Stop arrival detection

- ARRIVED: the bus is inside the stop radius (default 50 m) and its speed is below 5 km/h for 15 seconds.
- DEPARTED: the bus is outside the radius for 10 seconds, or its speed is above 10 km/h after arriving.

- A manual arrival (driver tap) is recorded with method MANUAL. A manual arrival overrides automatic detection for that stop only.
- Each event is stored in stop_events and used for the trip summary and for ETA.

17.5 ETA

1. Project the bus's current position onto the route line. Use the distance along the line from the route start.
2. For each remaining stop, compute the remaining distance along the line.
3. Speed estimate: the median of the bus's moving speeds (above 3 km/h) over the last 5 minutes, clamped between 10 and 60 km/h (defaults). If there is no moving sample in the window, use 20 km/h.
4. ETA = remaining distance divided by speed estimate, rounded to the nearest minute.
5. Display rules: under 60 seconds shows Arriving. A stop the bus has passed shows Departed. ETAs are recalculated on every snapshot.

The MVP does not snap the displayed marker to the road. Phase 2 adds snapping (section 40).

17.6 Stale and unavailable display

- STALE: the last known position is shown in grey, with its age in seconds.
- UNAVAILABLE: the last known position is shown in grey with the time of last contact. No ETA is shown.
- NOT_RUNNING: no marker is shown. The route card shows the next scheduled trip.
- A position is never shown as live when its age is over 60 seconds.

18. Attendance

18.1 Boarding QR code

- The driver screen shows a QR code that changes every 30 seconds.
- The token contains the bus ID, trip ID, and a 30-second time slot, signed with HMAC-SHA256 using the current attendance key (key_version).
- The server accepts the current slot and the previous slot, which covers 30 seconds of clock and scan delay.

- The attendance key rotates monthly. The previous key stays valid for 7 days after rotation.
- The QR code appears only during an ACTIVE trip and stops when the trip ends. The student app never displays it.

18.2 Validating a student scan

The server checks in this order. The first failure stops the check and returns its code:

1. Session is valid (AUTH_REQUIRED).
2. Token decodes and its HMAC is valid (QR_INVALID).
3. Time slot is within the window (QR_EXPIRED).
4. Trip is ACTIVE for the bus in the token (TRIP_NOT_ACTIVE).
5. The student is expected on this trip: on the route for the trip type and date, and not on leave (NOT_ON_ROUTE).
6. No BOARDED or LATE_BOARDED record exists for this trip and student (DUPLICATE_BOARDING).
7. The boarding window is open, or the trip has a late check-in permission (BOARDING_WINDOW_CLOSED).
8. Write the record: status BOARDED, method QR, boarding_stop_id set to the student's boarding stop.
9. Return the confirmation with the bus number and time.

18.3 Manual check-in (driver)

- The driver searches the expected list by name or roll number and taps the student.
- A student who is not on the expected list can be recorded as FLAGGED_GUEST, with a required reason. Flagged guests are not counted as passengers.
- A check-in after the window closes is recorded as LATE_BOARDED. A reason is required.
- Every manual check-in is logged with the driver and time.

18.4 Status transitions

From	To	Trigger	Notes
(none)	EXPECTED	Shift start creates the list	Built from route assignments, minus leave
EXPECTED	BOARDED	QR scan or driver check-in inside the window	Method recorded
EXPECTED	LATE_BOARDED	Driver check-in after the window, with reason	Counted as a passenger
EXPECTED	ON_LEAVE	Leave record created before the cutoff	Never changes automatically afterwards
EXPECTED	ABSENT	Boarding cutoff passes with no boarding	Set by the boarding-cutoff job
ABSENT	LATE_BOARDED	Driver check-in with reason after the cutoff	Absence alert, if raised, closes with status BOARDED_LATE
BOARDED, LATE_BOARDED	Changed only by coordinator correction	Requires a reason	Logged in the audit log

18.5 Boarding window

- Opens 10 minutes before the trip's scheduled departure (default).
- Closes 15 minutes after the scheduled departure (default).
- For a late-started trip, the window is measured from the actual departure.

18.6 Morning-present, afternoon-absent rule

This rule creates an absence alert. It is evaluated by the absence-evaluator job (section 25).

- Applies to a student with EXPECTED records on both the MORNING and AFTERNOON trips of the same service date.
- Trigger: the afternoon boarding cutoff has passed.
- Condition: the morning record is BOARDED or LATE_BOARDED, and the afternoon record is ABSENT. A student on leave for the afternoon never triggers the alert.
- Action: create an absence_alerts row and notify the primary faculty member for the afternoon bus. If there is no primary, notify all assigned faculty. If there is none, notify the coordinator.
- Once per student per service date.
- If the student boards the afternoon trip late, before the alert is acknowledged, the alert closes with status BOARDED_LATE and the faculty member is told.
- The coordinator's daily summary (section 21) lists every alert for the day.
- A student who missed the morning trip does not trigger this rule. Other patterns are a phase 2 option (section 40).

18.7 Leave

- A student can mark leave for a date and trip type until 30 minutes before the trip's scheduled departure.
- Faculty can mark leave for students on their bus until the boarding cutoff.
- Leave sets the expected record to ON_LEAVE immediately.
- Cancelling leave is allowed until the cutoff. It reverts the record to EXPECTED, but only if no boarding has occurred.
- Leave records store a category only (Personal, Academic, Other).

18.8 RFID check-in (phase 2)

- The coordinator registers a card for a student. Only a hash of the card UID is stored.
- An RFID_TAP event matches the hash to a student, then runs the same checks as steps 4 to 9 in section 18.2.
- Card loss is handled by deactivating the card; the student gets a new one.

18.9 Visibility and retention

- A student sees their own attendance.
- Faculty see attendance for trips on their buses.
- The coordinator sees all attendance. Management sees counts only.
- Attendance exports are logged. Retention follows section 29.

18.10 Acceptance criteria

- AC-ATT-01: Given a QR code from 30 seconds ago, the scan is accepted. Given one from 65 seconds ago, the scan returns QR_EXPIRED.
- AC-ATT-02: Given a second scan for the same trip, the response is DUPLICATE_BOARDING and no second record is written.
- AC-ATT-03: Given a student boarded the morning trip and did not board the afternoon trip by cutoff, the faculty alert is created within 5 minutes of the cutoff.
- AC-ATT-04: Given the student is on leave for the afternoon, no alert is created.

19. Emergency (SOS)

19.1 Triggers

Trigger	Sent by	Input	Notes
DRIVER_APP	Driver app	Hold for 2 seconds, then category and optional message	Category: Emergency, Medical, Breakdown, Other
HW_BUTTON	Hardware button	Button held for 300 ms (debounced)	No category. Category recorded as UNKNOWN
STUDENT_APP	Student app	Hold for 2 seconds, then optional message	Sends the bus and trip only. Student location is never sent (R-19)
FACULTY_APP	Faculty app	Hold for 2 seconds, then category	Same as driver
SECURITY_APP	Security app	Hold for 2 seconds	Same as driver

The hold requirement prevents accidental activation. The app shows a countdown while the button is held.

19.2 Creating or joining an incident

- An incident is created or joined within 1 second of the request reaching the server.
- If an OPEN incident exists for the same bus and the new trigger arrives within 60 seconds of the first trigger (default), the new trigger is recorded as a JOINED event on that incident.
- Otherwise a new incident is created.
- Two buses triggering at the same moment always produce two incidents. An alert is never suppressed because another alert has the same time or place.

19.3 Recipients by escalation tier

Tier	Starts	Recipients	Channels
0	Immediately	Faculty assigned to the bus, security staff on duty, coordinators on the duty list, and the sender (confirmation)	Push (high priority), in-app full-screen alert with sound
1	2 minutes with no acknowledgement	All coordinators, and the bus escalation contacts	Push, SMS (DLT template), in-app
2	5 minutes with no acknowledgement	Super admins, the security office number, and the college emergency desk email	Push, SMS, email
3	10 minutes with no acknowledgement	The college emergency contact list, by SMS, repeated every 5 minutes until resolved	SMS, push

The MVP sends no automatic voice calls. Tier 3 therefore needs a person to telephone the emergency contact list. Automatic calling is a phase 2 item (section 40). The on-duty lists and escalation contacts are maintained in configuration (section 35).

19.4 Stopping escalation

- Escalation stops for the incident when any FAC, SEC, or COR user marks it RESPONDING.
- Acknowledgement by any recipient stops that recipient's reminders only.
- The sos-escalator job checks open incidents every 15 seconds.

19.5 Delivery requirements

- Android: a notification channel named SOS with high importance and full-screen intent.
- iOS: time-sensitive interruption level. Critical Alerts need an Apple entitlement and are out of MVP scope unless approved (section 41).
- In-app: an alarm sound plays while an incident is open and unacknowledged. Volume follows the device alarm setting where the platform allows.
- Each recipient's delivery state is recorded: QUEUED, SENT, DELIVERED (where the provider gives a receipt), or FAILED.
- If an incident has zero delivered recipients after 60 seconds, a Severity 1 alert is raised (section 22).

19.6 Cancellation

- The user who triggered the incident can cancel within 30 seconds (default).
- A reason is required: Accidental press, Resolved before sending, or Other.
- After any acknowledgement, cancellation needs a coordinator.
- A cancelled incident has status CANCELLED. All recipients are told, and the event is logged.

19.7 Resolution

- A responder (FAC, SEC, or COR) marks the incident RESOLVED with a note of up to 500 characters. The note is required.
- All recipients are notified.
- The incident timeline can be exported as CSV (section 23).

19.8 Device delays

- The hardware sends triggered_at. The server records received_at.

- If the difference is over 60 seconds, the incident is labelled DEVICE_DELAYED and shows the delay.
- The device resends every 5 seconds until it receives a response with incident_id (section 12).

19.9 Drills

- A coordinator can run a drill, flagged is_drill. Recipients see a TEST label.
- Drills never escalate beyond tier 0.
- Drill records are kept for one year.
- A monthly drill is recommended.

19.10 Acceptance criteria

- AC-SOS-01: Given an SOS from the driver app, the incident exists within 1 second, and tier 0 push is sent within 5 seconds at the 95th percentile.
- AC-SOS-02: Given two triggers from the same bus 20 seconds apart, one incident exists with two trigger events.
- AC-SOS-03: Given two buses trigger in the same second, two incidents exist.
- AC-SOS-04: Given no acknowledgement for 2 minutes, tier 1 starts and coordinators receive push and SMS.
- AC-SOS-05: Given a faculty member has acknowledged, the trigger user's cancellation is refused with a message that a coordinator must cancel.

20. Stop requests

20.1 Submitting a request (student)

1. The student taps Request a stop, places a pin on the map, and enters a proposed name (up to 60 characters).
2. The route is taken from the student's current route assignment.
3. The server runs the checks below. A failed check returns a message and creates nothing.
    - Corridor check: the pin must be within 300 metres (default) of the route geometry. Otherwise: REJECT_OUTSIDE_ROUTE.

    - Duplicate of an active stop: within 100 metres (default) of an approved stop on this route. The response names the existing stop, and no request is created.
    - Duplicate of a pending request: within 100 metres of a PENDING request on this route. The student is recorded as a vote on that request.
    - Limit: one new request per student per 7 days.
4. A valid request is created with status PENDING.

20.2 Votes

- Each student can vote once per request.
- Votes are shown to the coordinator as a count. Votes never approve a request automatically.

20.3 Review (coordinator)

- The review screen shows the pin on the route map, the proposed name, the vote count, and the distance from the route line.
- Approve: edit the name and pin, choose the position in the route sequence from a dropdown of existing stops, and set effective_from (default: the next service day).
- Reject: a reason is required, chosen from Outside service area, Too close to an existing stop, Unsafe location, or Other (with text).
- Merge: combine this request into another pending request. Votes are added together.

20.4 On approval

- Creates a stop record and inserts it into route_stops at the chosen sequence, valid from effective_from.
- Existing trips before effective_from are unchanged. The next snapshot includes the new stop.
- Notifies the requester (category ADMIN). Notifies route students (category ADMIN, route change, default on).

20.5 Rules

- A request never edits an existing stop (R-13).
- Deactivating a stop is a separate coordinator action with an effective date. Students whose boarding stop is that stop receive a notification to choose a new stop in the app before that date. If they do not choose, the coordinator's default applies.

20.6 Acceptance criteria

- AC-STP-01: Given a pin 400 metres from the route, the request is rejected with REJECT_OUTSIDE_ROUTE.
- AC-STP-02: Given a pin 40 metres from an active stop, no request is created and the existing stop is named.
- AC-STP-03: Given approval with an effective date of next Monday, the stop is absent from trips before that date and present from that date.

21. Notifications

21.1 Categories

Category	Examples	Default channels	Can be turned off	Quiet hours apply
SAFETY	SOS and SOS status changes, bus unavailable during a shift (faculty)	Push, in-app. SMS for SOS	No	No
ATTENDANCE	Boarding confirmation, absence alert, leave confirmation	Push, in-app	Boarding confirmation: yes. Absence alert (faculty): no	Yes, except absence alerts
TRIP	Trip started, approaching stop, late start, delay, cancellation	Push, in-app	Yes	Yes
ADMIN	Stop request decision, announcements, route change, expiry reminders	Push, in-app, email	Yes	Yes
SYSTEM	Sign-in from a new device, consent updates, data request updates	Email, in-app	Consent and security: no. Others: yes	Yes

21.2 Delivery

- Each notification row is written in the same database transaction as the event that causes it (outbox pattern). A notification is never lost when a provider is down.
- The dispatcher sends each delivery on its channel. Failed deliveries are retried after 1 minute, 5 minutes, 30 minutes, and 2 hours. After 6 attempts a delivery is dead-lettered and an operational alert is raised (section 22).

- Each delivery records its status: QUEUED, SENT, DELIVERED (where a receipt is available), FAILED, or SUPPRESSED (quiet hours or preference).

21.3 Limits

- Non-safety push: at most 6 per user per hour.
- Notifications of the same category within 2 minutes are grouped into one message.
- SAFETY notifications are never rate-limited.

21.4 Quiet hours

- Default: 22:00 to 06:00 campus time. Configurable (section 35).
- During quiet hours, non-safety push is suppressed and the item is kept in the in-app notification center.
- SAFETY ignores quiet hours.

21.5 SMS rules

- Only DLT-registered templates with an approved sender ID are used.
- Message text matches a registered template exactly, with variables filled in.
- SMS never contains coordinates. It can contain a link to the web console, which requires sign-in.
- Every SMS send is logged with the template ID.

21.6 Email rules

- Sent through Amazon SES.
- Non-safety email includes an unsubscribe link that changes the preference for that category.
- Safety email contains no unsubscribe link.

21.7 Preferences

- Preferences are set per category and channel in Settings.
- Defaults are set in the table above.
- SAFETY preferences cannot be changed.

21.8 In-app notification center

- Lists notifications newest first, with unread count and mark-all-read.
- Notifications are kept for 90 days.

21.9 Coordinator daily summary

Sent at 08:00 campus time (configurable). Covers: SOS incidents in the last 24 hours, absence alerts, devices delayed or offline, stop requests pending, trips late or cancelled, and documents expiring within 30 days.

21.10 Templates

- Stored in the sms_templates table (SMS) and in template files (push and email).
- Variables are named, for example bus_number, trip_time, and stop_name.
- Strings are externalised so that another language can be added without code changes.

22. Operational alerts

Severity definitions: Severity 1 means a service or safety impact now, and pages the on-call engineer at any hour. Severity 2 means degraded service, and needs a response within one hour during operating hours. Severity 3 means a fix is needed by the next working day.

Alert	Condition	Severity	Recipients	Automatic action
Fleet silence	More than 20 percent of buses on active shifts have no valid data for 3 minutes	1	Technician, coordinator, on-call engineer	None
Bus unavailable	Active shift with no fresh source for 2 minutes	2	Coordinator and faculty	Bus state UNAVAILABLE (section 14)
Device offline	Device health OFFLINE (section 11.6)	3, or 2 during an active shift	Technician	Ticket

Alert	Condition	Severity	Recipients	Automatic action
Source flapping	More than 6 source switches in 10 minutes	3	Technician	Hold the current source for 5 minutes (section 13.5)
Authentication failures	More than 50 failed device authentications in 5 minutes for one device	2	Security reviewer, technician	Suggests revocation. Never revokes automatically
Rejection spike	More than 20 percent of one device's samples rejected in 10 minutes	3	Technician	None
GPS frozen	Speed above 5 km/h reported while the position is unchanged for 5 minutes	3	Technician	Ticket
Low vehicle battery	LOW_VEHICLE_BATTERY event	3	Technician, coordinator	Ticket
Certificate expiry, under 30 days	Scheduled check	2	Technician, on-call	None
Certificate expiry, under 7 days	Scheduled check	1	Technician, on-call	None
API errors	5xx rate above 1 percent for 5 minutes	1	On-call	None
Database failover	Failover event from the managed database	2	On-call	None
SOS not delivered	Incident with zero delivered recipients after 60 seconds	1	On-call, coordinator	Retries every channel once
Backup stale	Last successful backup older than 25 hours	2	On-call	None

Alert	Condition	Severity	Recipients	Automatic action
Job backlog	Queue depth above 1,000, or oldest job older than 2 minutes	2	On-call	None
Document expiry	Bus document expires in 30 or 7 days	3	Coordinator	Reminder notification
Unlinked device	Device sends data for more than 24 hours with no bus link	3	Coordinator	None

23. Reports and exports

Report	Contents	Roles	Format
Attendance by trip	Date, route, bus, trip type, expected, boarded, late, absent, on leave	COR, FAC (own buses), MGT (counts only)	CSV
Absence alerts	Date, student identifier, morning status, afternoon status, alert time, outcome	COR, FAC (own buses)	CSV
Trip history	Distance, duration, average speed, stops, boardings, failovers, SOS count, longest gap	COR, FAC (own buses), MGT (aggregate)	CSV
Source usage	Per bus per day: percent on HW, PHONE, and no source. Switches by reason	COR, TEC	CSV
Device uptime	Per device per day: time ONLINE, DELAYED, OFFLINE. Battery and signal averages	TEC, COR	CSV
SOS incidents	Timeline, recipients, acknowledgements, resolution	COR, SEC (own view), SAD	CSV

Report	Contents	Roles	Format
Stop requests	Requests, votes, decisions, effective dates	COR	CSV
Users	Role, status, route, bus and faculty assignments	COR, SAD	CSV (no sensitive columns)
Audit log	Action, actor, entity, time, old and new values	SAD. COR read-only	CSV
Data requests	Type, status, due date, completion	COR, SAD	CSV

Rules:

- Exports are generated by a background job and stored in S3. The download link is a signed URL valid for 24 hours.
- An export is limited to 100,000 rows. Larger requests are split by date range.
- Each export is written to the audit log with its filters and row count.
- Columns are limited by role. Student identifiers are hidden from MGT.
- PDF output is a phase 2 item.

24. Admin console: global behaviour

- Top bar: role switcher, search across buses, routes, users, and devices, notifications, and sign-out.
- Status strip, visible on every page: number of live buses, stale, unavailable, and open SOS incidents. An open SOS is shown in red and cannot be dismissed.
- Idle timeout: 30 minutes for COR, TEC, MGT, and SAD, with a warning at 25 minutes. 15 minutes for SEC.
- Destructive actions need a confirmation step. Unsaved changes trigger a warning before navigation.
- Keyboard navigation and screen reader labels on every control (WCAG 2.1 AA, section 30).
- Every list page has an empty state, a loading state, and an error state with a retry action.

- Page map: Dashboard, Live map, Buses, Devices, Routes, Stops, Timetables, Assignments, Users, Stop requests, SOS, Alerts, Source log, Attendance, Announcements, Reports, Audit log, Settings.

25. Background jobs

All jobs run in the worker process. Each job is idempotent, so a retry never creates duplicate data. Failed jobs retry with backoff and are dead-lettered after 6 attempts (section 22 raises the alert).

Job	Schedule	Purpose	Failure handling
source-selector	Every 5 seconds per active bus, and on each accepted sample	Source selection (section 13)	Retry. Alert if lag exceeds 30 seconds
stale-checker	Every 30 seconds	Stale, unavailable, and device health states	Retry
shift-auto-end	Every 60 seconds	Idle prompts and auto-end (section 15.4)	Retry. Idempotent per shift
trip-generator	Daily at 00:30 campus time	Creates trips for the next 7 days (section 15.5)	Retry. Alert on failure
late-start-check	Every 60 seconds in service hours	TRIP_LATE_START (section 15.6)	Retry
attendance-expected-builder	On trip start	Creates EXPECTED records	Unique key prevents duplicates
boarding-cutoff	Every minute	Sets ABSENT after the cutoff (section 18.4)	Idempotent
absence-evaluator	At each afternoon cutoff, plus every 5 minutes as a safety net	Morning-present, afternoon-absent rule (section 18.6)	Once per student per date

Job	Schedule	Purpose	Failure handling
sos-escalator	Every 15 seconds	Escalation tiers (section 19.3)	Must not skip. Alert if lag exceeds 30 seconds
notification-dispatcher	Continuous	Sends queued notifications (section 21.2)	Retries per schedule
device-health-checker	Every minute	ONLINE, DELAYED, OFFLINE (section 11.6)	Retry
cert-and-token-check	Daily	Certificate and device token expiry	Alert (section 22)
retention-purger	Daily at 02:00	Drops old partitions and deletes records past retention (section 29)	Logged. Can be rerun manually
partition-manager	Daily	Creates partitions 7 days ahead	Alert if a partition is missing
export-generator	On request	Report files (section 23)	Retry twice. Notify requester on failure
coordinator-summary	08:00 campus time (configurable)	Daily summary (section 21.9)	Retry
backup-verifier	Daily	Confirms backup freshness and a sample restore	Alert (section 22)

26. API catalogue

All paths are under /api/v1. Roles use the codes in section 4. Scope rules in section 5 apply to every read. Ingestion endpoints are served from the dedicated ingest hostname (section 34).

26.1 Authentication and profile

Method and path	Roles	Purpose
POST /auth/sso/start	Public	Start college SSO sign-in
GET /auth/sso/callback	Public	SSO callback, creates session
POST /auth/otp/request	DRV	Send a sign-in code
POST /auth/otp/verify	DRV	Verify the code, create session
POST /auth/refresh	All signed in	Rotate refresh token
POST /auth/logout	All signed in	End the current session
GET /auth/sessions	All signed in	List own sessions
DELETE /auth/sessions/{id}	All signed in	Revoke one of own sessions
POST /auth/mfa/enrol	Admin roles	Start TOTP enrolment
POST /auth/mfa/verify	Admin roles	Verify a TOTP code
GET /me	All signed in	Profile, roles, permissions summary
PATCH /me	All signed in	Update name and phone (email is changed by coordinator only)
GET /me/notification-preferences	All signed in	Read preferences
PUT /me/notification-preferences	All signed in	Update preferences (SAFETY cannot be changed)
POST /me/data-requests	All signed in	Export, correction, deletion, or consent withdrawal request
GET /me/data-requests	All signed in	Status of own requests

26.2 Live tracking

Method and path	Roles	Purpose
GET /snapshot?route_id=	STU, FAC, DRV, SEC, COR, SAD	Bus states for one route (section 17.3)
GET /fleet/live	SEC, COR, SAD, MGT (aggregate only)	All active buses
GET /buses/{id}	Per scope	Bus detail
GET /buses/{id}/track?minutes=	COR, SAD	Both sources for comparison (section 13.8)
GET /routes/{id}/geometry	STU, FAC, DRV, COR	Route line and stops with version

26.3 Ingestion (device and driver)

Method and path	Sender	Purpose
POST /telemetry	Device	Samples and events (section 11.3)
POST /heartbeat	Device	Health report (section 11.6)
POST /devices/token/rotate	Device	Receive rotated token (section 11.1)
POST /driver/locations	Driver session	Phone samples (section 11.7)
POST /driver/status	Driver session	Phone health (section 11.7)

26.4 Fleet, devices, and firmware

Method and path	Roles	Purpose
GET, POST /buses	COR, SAD create. TEC and MGT read	Bus list and create
GET, PATCH /buses/{id}	COR, SAD write. TEC and MGT read	Bus detail and update

Method and path	Roles	Purpose
POST /devices/batch	TEC	Provision a batch of devices (section 16.5)
GET /devices, GET /devices/{id}	TEC, COR	Device registry and detail
POST /devices/{id}/tokens	TEC	Request token rotation
POST /devices/{id}/revoke	COR, SAD	Revoke token (section 11.1)
POST /device-assignments	COR	Link device to bus
POST /device-assignments/{id}/close	COR	Close a link with a date
POST /firmware-releases	TEC	Register a firmware release
POST /firmware-rollouts	TEC	Start or pause a rollout (section 33)
POST /tickets	DRV, TEC, COR	Open a maintenance ticket
GET, PATCH /tickets/{id}	TEC, COR. DRV reads own	Ticket list, update, close

26.5 Routes, stops, timetables

Method and path	Roles	Purpose
GET, POST /routes	COR writes. Others read per scope	Route list and create
GET, PATCH /routes/{id}	COR writes	Route detail and update
PUT /routes/{id}/stops	COR	Set the ordered stops with validity dates
PUT /routes/{id}/geometry	COR	Replace the route line (creates a new version)
GET, POST /stops	COR	Stop library
PATCH /stops/{id}	COR	Edit or deactivate a stop (with effective date)

Method and path	Roles	Purpose
GET, POST /timetables	COR	Timetables per route and trip type
POST /route-bus-assignments	COR	Assign buses to a route for a trip type and weekday mask

26.6 Assignments and users

Method and path	Roles	Purpose
POST /roster, POST /roster/{id}/close	COR	Driver roster
POST /faculty-assignments, POST /faculty-assignments/{id}/close	COR	Faculty to bus
POST /student-routes, POST /student-routes/{id}/close	COR	Student to route
POST /student-routes/{id}/boarding-stop	STU (own), when notified	Choose a new boarding stop
GET, POST /users	COR creates. Others read per scope	User list and invitation
POST /users/import	COR	CSV import, dry run then commit (section 36)
PATCH /users/{id}	COR	Status change, deactivation with end date
POST /users/{id}/roles	SAD	Grant or revoke a role

26.7 Shifts, trips, and attendance

Method and path	Roles	Purpose
POST /shifts	DRV	Start a shift (after pre-trip check)
GET /shifts/current	DRV	Resume the current shift

Method and path	Roles	Purpose
POST /shifts/{id}/end	DRV (own), COR	End a shift
POST /shifts/{id}/mode	DRV (own), COR	Change mode with reason
POST /shifts/{id}/prompt-response	DRV (own)	Answer the idle prompt
GET /trips	Per scope	Trips by date, route, or bus
POST /trips/{id}/cancel	COR	Cancel a trip with reason
GET /trips/{id}/summary	COR, FAC (own), MGT (aggregate)	Trip summary (section 15.7)
GET /trips/{id}/passengers	DRV (own shift), FAC (own), COR	Expected list with status
GET /trips/{id}/qr-token	DRV (own shift)	Current QR payload (section 18.1)
POST /attendance/scan	STU	Student QR scan (section 18.2)
POST /attendance/manual	DRV (own shift), COR	Manual check-in (section 18.3)
POST /attendance/{id}/correct	COR	Correct a status with reason
GET /attendance/me	STU	Own attendance history
POST /leave	STU, FAC	Create leave
DELETE /leave/{id}	Creator, before cutoff	Cancel leave
GET /absence-alerts	FAC (own), COR	Alerts list
POST /absence-alerts/{id}/ack	FAC (own), COR	Acknowledge with note

26.8 Emergencies

Method and path	Roles	Purpose
POST /sos	All except MGT and TEC	Trigger an incident (section 19)

Method and path	Roles	Purpose
GET /sos	SEC, COR, SAD. FAC for own buses	Incident list, filter by status
GET /sos/{id}	Per scope	Incident with timeline
POST /sos/{id}/ack	FAC, SEC, COR	Acknowledge
POST /sos/{id}/responding	FAC, SEC, COR	Mark responding (stops escalation)
POST /sos/{id}/cancel	Trigger user (30 s), COR	Cancel with reason
POST /sos/{id}/notes	FAC, SEC, COR	Add a note
POST /sos/{id}/resolve	FAC, SEC, COR	Resolve with note
POST /sos/drills	COR	Run a drill

26.9 Stop requests, notifications, announcements, exports, settings

Method and path	Roles	Purpose
POST /stop-requests	STU	Submit a request (section 20.1)
POST /stop-requests/{id}/vote	STU	Vote on a pending request
GET /stop-requests	COR. STU reads own	List requests
POST /stop-requests/{id}/approve, /reject, /merge	COR	Review actions (section 20.3)
GET /notifications	All signed in	Own notifications
POST /notifications/{id}/read, POST /notifications/read-all	All signed in	Mark read
GET, POST /announcements	COR writes. Others read per scope	Announcements

Method and path	Roles	Purpose
POST /exports	COR. FAC (attendance), TEC (device), MGT (aggregate)	Start an export (section 23)
GET /exports/{id}	Requester	Export status and signed link
GET /audit-logs	SAD. COR read-only	Audit log search
GET /settings, PUT /settings/{key}	COR reads. SAD writes, with reason	Thresholds and settings (section 35)
GET, PUT /feature-flags/{key}	SAD	Feature flags
GET /health	Public	Liveness only, no data
GET /health/ready	Internal network only	Readiness, checks database access

27. API conventions and errors

- Base URLs: api and ingest hostnames under the college domain, served over HTTPS only.
- Content type: application/json; charset=utf-8.
- Timestamps: UTC, ISO 8601, with the Z suffix. Clients convert to IST for display.
- Pagination: cursor-based. Parameters limit (default 50, maximum 200) and cursor. Responses have the shape data and next_cursor.
- Idempotency: state-changing requests from the apps and console require an Idempotency-Key header (UUID). The server stores the response for 24 hours. A repeat with the same key returns the stored response. A repeat with a different body returns IDEMPOTENCY_CONFLICT.
- Request ID: every response includes X-Request-Id. Support tickets quote this value.
- Out-of-scope resources return 404, not 403, so the response does not reveal that the resource exists.
- Rate limit headers on every response: RateLimit-Limit, RateLimit-Remaining, RateLimit-Reset.
- Deprecated fields and endpoints carry Deprecation and Sunset headers (section 11.9).

Error envelope:

```
{
  "error": {
    "code": "DUPLICATE_BOARDING",
    "message": "You are already boarded on this trip.",
    "details": {"boarded_at": "2026-10-08T07:42:00Z"},
    "request_id": "req_01HZX3K9Q2"
  }
}
```

Code	HTTP	Meaning
AUTH_REQUIRED	401	No valid session
SESSION_EXPIRED	401	Refresh failed or session revoked
DEVICE_REVOKED	401	Device token revoked
FORBIDDEN	403	Role not permitted for this action
DEVICE_INACTIVE	403	Device retired or revoked
NOT_ON_ROUTE	403	Student not on this trip
NOT_FOUND	404	Not found, or outside the caller's scope
VALIDATION_FAILED	400	Body does not match the schema. Field errors listed
QR_INVALID	422	QR token fails its signature check
QR_EXPIRED	422	QR token outside the current or previous slot
CONFLICT	409	State conflict, for example a duplicate CONFLICT sample
DUPLICATE_BOARDING	409	Already boarded on this trip
BOARDING_WINDOW_CLOSED	409	Outside the boarding window
SHIFT_NOT_ACTIVE	409	Shift is not active
BUS_ALREADY_ON_SHIFT	409	Another shift holds this bus
TRIP_NOT_ACTIVE	409	Trip is not active

Code	HTTP	Meaning
IDEMPOTENCY_CONFLICT	409	Same key used with a different body
PAYLOAD_TOO_LARGE	413	Body exceeds the endpoint limit
RATE_LIMITED	429	Rate limit exceeded. Retry-After set
SERVICE_UNAVAILABLE	503	Maintenance or overload. Retry-After set

28. Security requirements

Each item is a requirement for the MVP. The security test matrix (T-SEC, section 32) verifies them.

- Transport: HTTPS only, TLS 1.2 minimum and TLS 1.3 preferred. Plain HTTP requests are redirected or refused. HSTS is set on the web console. Certificates renew automatically (section 34).
- Authentication: as section 8. OTP codes are single-use. Failed attempts are counted and locked. Admin roles require TOTP.
- Authorisation: every endpoint checks role and scope. Data queries are filtered by assignment in the database layer, not only in the controller. An authorisation test covers every endpoint against every role.
- Input validation: every request is validated against its schema. Body size limits apply to every endpoint (section 11). SQL is parameterised only. CSV imports are limited to 5 MB and parsed in streaming mode.
- Spreadsheet safety: exported cells that start with =, +, -, or @ are prefixed with an apostrophe to block formula injection.
- Output: error responses never include stack traces, internal paths, or SQL. Responses include only the fields the caller's role needs.
- Data at rest: database and object storage are encrypted with a managed KMS key. Date of birth, guardian contact, and TOTP secrets are also encrypted at the field level.
- Secrets: held in AWS Secrets Manager. Never in code, images, or logs. Rotated every 90 days. The device-token key supports two active versions during rotation.
- Device credentials: per device, hashed, revocable (section 11.1). No shared keys.
- Firmware: signed and checksummed (section 33).
- Mobile storage: tokens and the device token are kept in platform secure storage only.

- Web headers: a Content Security Policy that blocks inline scripts from unknown sources, X-Frame-Options set to DENY, SameSite=Strict cookies, and a CSRF token on every write.
- Network: database and tasks sit in private subnets with no public address. Only the load balancer is public. Security groups allow only the ports each component needs.
- Access: least-privilege IAM roles. Root account protected with hardware MFA. CloudTrail enabled in every account.
- Rate limits and WAF: managed WAF rules on both hostnames, plus rate rules on the ingestion and sign-in endpoints.
- Dependencies: lockfiles committed. Dependency scanning on every build. Critical vulnerabilities patched within 7 days, high within 30 days.
- Penetration test: before launch and once a year. Findings of high or critical severity block release.
- Audit: the audit log is append-only at the database level (R-09).

29. Privacy, consent, and retention

29.1 Principles

- Collect only what each feature needs (data minimisation).
- Students' own location is never collected in the MVP (R-19).
- Driver phone location is collected only during an active shift, with a visible indicator (R-18).
- No third-party analytics or advertising SDK receives location or personal data. Crash reporting strips personal data before sending.
- Students and drivers can see what the app collects in the Privacy screen.

29.2 Consent and notices

- Privacy notice versions are stored in notice_versions. Each user sees the current version at sign-in and when it changes.
- Consent is recorded per purpose in consents: app use and account, driver location during shifts, attendance records, and notifications.
- Students under 18: age is taken from date_of_birth. If date_of_birth is missing, the student is treated as under 18 until the coordinator confirms the date.

- Parental consent for under-18 students: a verification link is sent to the guardian contact. The guardian confirms by entering a one-time code sent to that contact. The consent record stores the guardian's identifier and the time. The college's counsel confirms that this method meets the Rules before launch.
- Until guardian consent is recorded, an under-18 student sees only the notice and the SOS button. No attendance, no notifications other than SAFETY, and no bus positions are shown. SOS remains available. The counsel confirms this position.
- Withdrawal of consent is available in Profile. Withdrawal stops the matching processing. It does not remove SOS or audit records.

29.3 Personal data requests

- Types: export, correction, deletion, and withdrawal of consent. Requests are recorded in data_requests with a due date. The internal target is 30 days.
- Identity: a signed-in user requests their own data. A guardian request is verified against the guardian contact.
- Export: a JSON file and a CSV file, delivered as a signed link valid for 24 hours.
- Correction: name, phone, and department can be corrected by the user. Other corrections are made by the coordinator.
- Deletion: personal fields are removed. Attendance and location records are deleted. SOS records and audit records are kept in anonymised form where law or policy requires them, and the user is shown what was kept and why.

29.4 Retention (defaults)

The college's counsel and policy owner set the final values.

Data	Default retention	At end of retention
Location samples, hardware and phone	90 days	Partition dropped
Device heartbeats	90 days	Partition dropped
Device events	1 year	Deleted
Trip summaries	2 years	Deleted
Attendance records	2 years, or the academic-year policy	Deleted

Data	Default retention	At end of retention
Absence alerts	1 year	Deleted
SOS incidents and timelines	3 years	Personal fields anonymised. Aggregate counts kept
Stop requests	2 years	Deleted
In-app notifications	90 days	Deleted
Notification delivery logs	90 days	Deleted
Audit logs	2 years	Deleted
Exports	7 days	Deleted from storage
Application logs	30 days	Deleted. Contain no personal data
Accounts	Until deactivation plus 90 days	Personal fields deleted. Pseudonymous ID kept for historical reports

29.5 Access to location history

- Only COR and SAD can view location history (R-17).
- Exporting location history requires SAD approval and is logged.
- Every access to location history is written to the audit log.

29.6 Breach response

1. Detect: through alerts (section 22) or a user report.
2. Contain: rotate affected secrets, revoke affected sessions and tokens, and disable the affected feature if needed.
3. Assess: which data, whose data, and what risk.
4. Notify: the Data Protection Board and affected individuals within the timelines in the Rules. The counsel confirms the timelines. The internal target is 72 hours from confirmation.
5. Record: an incident log entry with decisions and times.

29.7 Drivers

- Drivers receive a notice explaining what is collected, when, and who can see it.
- The app shows the driver the status of location sharing at all times during a shift.
- Driver location follows the same retention periods as student-related location data.

30. Non-functional requirements

Area	Requirement	How it is measured
API availability	99.5 percent per month	External probes every 30 seconds
Ingestion availability	99.9 percent per month target. Tracked from launch	Ingestion success rate
Position freshness	95th percentile of position age at most 15 seconds for buses with a fresh source	Time from recorded_at to snapshot
SOS latency	Incident created within 1 second. Tier 0 push sent within 5 seconds. Both at 95th percentile	sos_events timestamps
API latency	Snapshot p95 at most 300 ms. Other reads p95 at most 500 ms. Writes p95 at most 800 ms	Metrics
Capacity	100 buses at launch, designed for 200. 5,000 registered users. 500 concurrent app users at peak	Load test (section 32)
Ingestion throughput	200 requests per second sustained	Load test
Recovery	Recovery point 15 minutes or less. Recovery time 1 hour or less	Restore test each quarter (section 34)
App cold start	3 seconds or less on a mid-range Android phone with 4 GB RAM	Device test
App size	40 MB or less (Android download). 60 MB or less (iOS app)	Build report

Area	Requirement	How it is measured
Driver battery	8 percent per hour or less of background tracking on the reference phone	Two-hour test (section 32)
Data use	Driver app 1 MB per hour or less in fallback mode. Student app 2 MB per day or less, typical	Network measurement
Offline behaviour	Driver app buffers 24 hours. Student app shows cached data for up to 10 minutes	Test cases T-OFF
Accessibility	WCAG 2.1 AA for the web console. Dynamic text up to 200 percent in the app. Labels on every control. Status is never shown by colour alone	Automated checks and device checks
Localisation	English at launch. All strings externalised. IST display. Dates shown as 18 Oct 2026	Review
Browser support	Last two versions of Chrome, Edge, Safari, and Firefox	Test matrix
Mobile OS	Android 10 and later. iOS 16 and later. Confirm against the devices the college actually has at kickoff	Store settings
Time sync	Server time from NTP. Devices sync from GNSS or the network	Clock-skew alerts

31. Observability and operations

31.1 Logging

- Structured JSON logs with timestamp, level, service, request_id, role, route, status, and duration.
- Logs contain device_id and bus_id. They never contain names, emails, phone numbers, coordinates, or tokens.
- Retention: 30 days.

31.2 Metrics

- Ingestion: requests and rejects by reason; samples accepted and rejected per minute.
- Source selection: transitions by reason code.
- Snapshot: latency and cache hit rate.
- SOS: creation-to-delivery latency, and delivery failures.
- Notifications: delivery rate per channel, and failures per provider.
- Jobs: queue depth, oldest job age, and failures.
- Devices: heartbeats per minute, and counts by health state.
- Security: authentication failures and lockouts.
- Database: connections, replication lag, and storage growth.

31.3 Tracing

OpenTelemetry traces across the API and workers, sampled at 10 percent. SOS and ingestion paths are sampled at 100 percent.

31.4 Dashboards

- Fleet health: buses live, stale, and unavailable; devices online, delayed, and offline.
- Ingestion: throughput, rejects, and latency.
- SOS: open incidents, escalation tiers, and delivery.
- Notifications: delivery and failures by channel.
- Platform: CPU, memory, database, and queue.
- Business: trips started against schedule, and boardings per trip.

31.5 External checks

Probes run every 30 seconds against the API, the ingest hostname, and the web console. Probes come from two providers. A probe failure is treated as an alert even when internal metrics look healthy.

31.6 Runbooks

Each runbook has: trigger, first five minutes, diagnosis steps, mitigation, communication template, and a post-incident review within five working days.

- RB-01 Fleet silence.
- RB-02 SOS not delivered.

- RB-03 API errors above threshold.
- RB-04 Database failover.
- RB-05 Device token compromise.
- RB-06 Certificate renewal failure.
- RB-07 Queue backlog.
- RB-08 Rolling back an API or app release.
- RB-09 Restore from backup.
- RB-10 Halting a firmware rollout.
- RB-11 Notification provider outage.
- RB-12 Personal data breach response (section 29.6).
- RB-13 Use of the emergency access account.

31.7 Incident communication

- Severity 1: the coordinator is told within 15 minutes, and updates follow every 30 minutes until resolved.
- A status page is maintained for users. In the MVP it is a static page.
- Severity 1 and 2 incidents get a written post-incident report.

31.8 On-call

- A rota of two engineers, covering the service hours in section 31.9 and all Severity 1 alerts at any hour.
- Severity 1 response target: 15 minutes.
- Escalation: an unresolved Severity 1 for 1 hour escalates to the super admin.

31.9 Service hours and change windows

- Service hours for live operations: 05:30 to 22:30 campus time.
- Releases avoid the morning and evening peaks. The peak windows are set in configuration.
- Database migrations run only in the change window, and only in the expand-then-contract pattern (section 33).

32. Testing

32.1 Test levels

Level	Scope	Tools	Gate
Unit	Domain logic: source selector, ETA, QR, validation rules, attendance rules, escalation	Vitest	At least 80 percent line coverage on domain modules
API integration	Full endpoints against PostgreSQL with PostGIS	Supertest, Testcontainers	Every endpoint has at least one success and one failure test
Contract	Replay of recorded messages from each firmware version	Test harness in the repository	Every build (section 11.9)
Web end-to-end	Coordinator core flows	Playwright	Every merge to main
Mobile end-to-end	Student, faculty, and driver core flows	Maestro, on real devices and a device farm	Every release candidate
Load	Ingestion and snapshot at target scale	k6	Before each minor release and before launch
Security	Authorisation matrix, injection, headers, secrets, dependencies	Custom matrix tests, OWASP ZAP baseline, dependency and secret scanners	Every merge (scanners) and before launch (matrix and penetration test)
Field	Hardware test bench and vehicle tests	Procedures in HW-SPEC section 16	Before firmware release
Pilot	Real users on three buses	Feedback form and logs	Exit criteria in section 39

32.2 Simulator

- The simulator in tools/simulator sends hardware and phone traffic. It can simulate any number of buses on routes with speed profiles, stop dwell, and a driver.
- Hardware behaviours: GNSS loss, cellular loss with buffered replay, duplicate delivery, out-of-order delivery, clock skew, reboot with a new boot_id, revoked token, and SOS press.
- Phone behaviours: permission revoked, app killed, mock flag set, and network drop.
- The simulator refuses to run against any production hostname. This check is hard-coded and tested.

32.3 Critical test cases

ID	Scenario	Expected result
T-ING-01	Valid batch of 50 samples	200. ack_seq is the 50th sequence. All samples stored
T-ING-02	Same batch sent twice	200 both times. No new rows on the second
T-ING-03	Same sequence, different content	Rejected with CONFLICT
T-ING-04	Timestamp 90 seconds in the future	Rejected with FUTURE_TIMESTAMP
T-ING-05	Coordinates 0, 0	Rejected with NULL_ISLAND
T-ING-06	Position jump of 200 km/h	Stored flagged. Rejected with IMPLAUSIBLE_JUMP
T-ING-07	Body of 9 KB	413. Device splits and resends
T-ING-08	Revoked token	401 DEVICE_REVOKED within 5 seconds
T-ING-09	Token rotation	New token activates on first use. Old token revoked. Pending token expires after 24 hours if unused
T-ING-10	Two-hour-old buffered samples	Accepted as BUFFERED. Never used for live state
T-SEL-01	Hardware stale for 60 seconds, phone fresh	Switches to PHONE with HW_STALE within 5 seconds of the threshold

ID	Scenario	Expected result
T-SEL-02	Hardware returns with three valid fixes in 30 seconds	Switches back to HW with HW_RECOVERED
T-SEL-03	Seven switches in 10 minutes	Holds current source for 5 minutes. Raises alert
T-SEL-04	Neither source fresh for 120 seconds	UNAVAILABLE. One alert
T-SEL-05	Manual HW mode, hardware dead	Stays STALE. No automatic switch
T-SHF-01	Start without Always location permission	Start blocked with fix steps
T-SHF-02	Second shift on the same bus	Blocked with BUS_ALREADY_ON_SHIFT
T-SHF-03	Driver not on the roster	Bus not listed. Direct request returns 403
T-SHF-04	No movement for 20 minutes with ignition off	Idle prompt. Auto-end after 10 more minutes without response
T-AUTH-01	Ten failed OTP attempts	Account locked for 15 minutes
T-AUTH-02	Account deactivated	All access removed within 60 seconds
T-AUTH-03	Old refresh token reused	Whole session family revoked
T-SEC-01	Authorisation matrix, every endpoint against every role	Results match section 5 and section 26
T-SEC-02	Student requests another route's snapshot	404
T-SEC-03	Export contains a cell starting with =	Cell is escaped
T-PRV-01	Under-18 student without consent	Sees only the notice and SOS
T-PRV-02	Deletion request	Attendance and location records removed. SOS records anonymised
T-NTF-01	SAFETY notification during quiet hours with preferences off	Delivered

ID	Scenario	Expected result
T-NTF-02	Seventh non-safety push in an hour	Suppressed and kept in the in-app center
T-OFF-01	Driver phone offline for 30 minutes	Samples buffered and flushed in order
T-OFF-02	Student app offline for 10 minutes	Cached data marked offline
T-LOAD-01	50 buses at 10-second intervals, 500 app users polling every 10 seconds	Snapshot p95 at most 300 ms. Ingest p95 at most 500 ms. No errors
T-LOAD-02	Five buses trigger SOS within 10 seconds	Five incidents. Tier 0 delivery within 5 seconds
T-FO-01	One API task killed under load	No failed ingestion after retries. Other task serves
T-FO-02	Database failover	Service recovers within the RTO. No data lost
T-FO-03	Certificate rotation rehearsal on staging	Test-bench module connects with the new chain
T-FO-04	Worker crashes during a job	Job retried. No duplicate records
T-MOB-01	Two hours of background tracking on each reference phone	At most 8 percent battery per hour
T-MOB-02	Battery optimisation enabled on Android	Pre-trip warning shown
T-A11Y-01	Automated accessibility scan of each console page	Zero critical violations

32.4 Test data

Test data is synthetic. Staging holds a simulated college with 50 buses, 2,000 accounts, and generated routes and stops. No real personal data is copied into any non-production environment.

32.5 Defect severity

- S1: blocks release. Includes any cross-role data exposure, SOS failure, or ingestion loss.
- S2: must have a documented workaround before release.
- S3: may ship, tracked in the backlog.

33. CI/CD, release, and firmware rollout

33.1 Branching

Trunk-based development. Branches live no longer than two days. Every pull request needs one review and passing checks.

33.2 Continuous integration

Each build runs: install from the lockfile, lint, type check, unit tests, contract tests, API integration tests, container image build, dependency scan, secret scan, static analysis, OpenAPI generation with a breaking-change check against the last released version, and the migration check.

33.3 Continuous deployment

- Staging deploys automatically when main changes.
- Production deployments need approval from an engineer other than the author, and from the super admin.
- Canary: one of two API tasks is updated first. The team watches error rate and latency for 15 minutes. Then the second task is updated.

33.4 Database migrations

- Expand, then contract. Additive changes first: new nullable columns and new tables. Deploy code that writes both old and new shapes. Backfill with a job. Remove old columns only in a later release.
- No release drops or renames a column or table in the same release that stops using it.
- Migrations run in the change window (section 31.9).

33.5 Feature flags

Risky features ship behind flags, off by default in production until verified: new selector thresholds, RFID, notification channels, and any new SOS tier.

33.6 Rollback

- API: redeploy the previous image within 10 minutes (RB-08).
- Database: no down-migrations in production. Fix forward.
- Mobile: store builds are not rolled back. A staged rollout is halted and a fix is shipped.

33.7 Mobile release

- Builds and submissions use EAS Build and EAS Submit.
- Android: staged rollout at 10 percent, then 50 percent, then 100 percent over three days. Crash-free sessions must stay at or above 99.5 percent.
- iOS: phased release over seven days.
- The API enforces a minimum supported app version. Below it, the app shows a forced-update screen.
- Over-the-air JavaScript updates are allowed for non-native fixes only. They are never used for permission or native-module changes.

33.8 Firmware release and rollout

- A release is registered with its version, SHA-256, and size.
- Devices download the image over HTTPS using a signed URL. They verify the signature and SHA-256 before installing, and keep the previous image for rollback.
- Rollout steps: a canary group of 5 buses, then 25 percent of the fleet, then the full fleet. Each step waits 48 hours with healthy metrics before the next.
- Pause triggers: FIRMWARE_FAILED above 2 percent of updates; any device going OFFLINE within 30 minutes of its update; a rise of more than 10 percentage points in sample rejections; or any SOS that was not delivered.
- Updates run only while the bus is parked with ignition off, unless the coordinator approves otherwise.
- No two buses on the same route are updated at the same time.
- Each firmware release must pass the contract tests against the backend release it targets.

34. Deployment and infrastructure

34.1 Hostnames

Hostname	Purpose	Notes
api.[college-domain]	Apps and console API	Standard certificate
ingest.[college-domain]	Device and driver-phone ingestion	Separate certificate chain, as the hardening plan requires. Short DNS TTL (60 seconds) so the target can be changed quickly
admin.[college-domain]	Admin web console	Standard certificate
status.[college-domain]	Public status page	Static

The device firmware stores the ingest hostname, never an IP address.

34.2 Region and data residency

An AWS region in India where every required service is available. Confirm at kickoff. All personal and location data is stored in India.

34.3 Compute

- API: ECS on Fargate, minimum 2 tasks across 2 availability zones. Scales between 2 and 6 tasks at 60 percent CPU.
- Workers: ECS on Fargate, minimum 2 tasks. Scales on queue depth.
- Web console: static files in S3, served through CloudFront.

34.4 Load balancing and health

- Application Load Balancer with HTTPS termination.
- Liveness check on /health every 10 seconds. Readiness check on /health/ready decides which tasks receive traffic. A task that cannot reach the database is removed from rotation but not restarted.
- Deregistration delay of 30 seconds, so in-flight ingestion completes.

- A separate listener rule routes the ingest hostname, behind WAF rate rules (section 28).

34.5 Database

- Amazon RDS for PostgreSQL 16 with PostGIS, Multi-AZ.
- Storage autoscaling enabled.
- Automated backups kept 7 days, with point-in-time recovery. Daily snapshots kept 30 days.
- Connection pooling through RDS Proxy.
- Partition management and retention run as jobs (section 25).

34.6 Object storage

Amazon S3: versioning on, default encryption with KMS, all public access blocked, lifecycle rule that deletes export files after 7 days.

34.7 Certificates

- AWS Certificate Manager for the load balancer, with automatic renewal.
- Expiry is monitored (section 22).
- The ingest certificate chain is tested on the EC200U-CN test module before launch (section 32, T-FO-03).

34.8 DNS

Amazon Route 53. The ingest hostname has a TTL of 60 seconds.

34.9 Secrets and configuration

- AWS Secrets Manager: database credentials, JWT signing keys, device-token key, SMS and push credentials, email credentials.
- AWS Systems Manager Parameter Store: non-secret configuration, including the values in section 35.

34.10 Infrastructure as code

- Terraform defines all resources.
- Dev, staging, and production run in separate AWS accounts.

- Production changes go through the pipeline only. Console changes are not allowed except through the emergency access procedure (RB-13), which is recorded.

34.11 Backups and restore

- Recovery targets are in section 30.
- A restore drill runs each quarter in a separate environment. Results are recorded, including the measured time.
- The backup verifier job runs daily (section 25).

34.12 Disaster recovery

- The MVP runs in one region. Multi-AZ protects against loss of a zone.
- For a regional outage, the documented procedure restores from backups into another India region. The expected time for this is about 4 hours, which is longer than the 1-hour target in section 30. The college accepts this risk, or upgrades to a standby region, as decision 41 records.

34.13 Cost controls

Budget alerts at 50, 80, and 100 percent of the monthly budget. A monthly cost review is held with the coordinator.

35. Configuration defaults

Every value below is held in configuration. Changes are made by the super admin, with a reason, and written to the audit log. The source of each value is the section that describes its behaviour.

Key	Default	Section
hw.stale_after_s	60	13.1
hw.min_satellites	4	13.1
hw.max_hdop	5.0	13.1
phone.max_accuracy_m	50	13.1
selector.cooldown_s	30	13.3

Key	Default	Section
selector.return_fixes	3	13.3
selector.return_window_s	30	13.3
selector.both_down_s	120	13.3
selector.flap_limit	6 switches in 600 s	13.5
selector.flap_hold_s	300	13.5
phone.standby_interval_s	30	13.6
phone.fallback_interval_s	10	13.6
hw.moving_interval_s	10	11.4
hw.stopped_interval_s	30	11.4
hw.parked_heartbeat_s	600	11.6
hw.heartbeat_moving_s	60	11.6
hw.gps_lost_after_s	30	12
hw.gps_lost_ticket_min	10	12
vehicle.battery_mv_12v	11800	12
vehicle.battery_mv_24v	23500	12
ingest.future_tolerance_s	60	11.5
ingest.buffered_after_min	10	11.5
ingest.stale_timestamp_days	7	11.5
ingest.jump_max_kmh	150	11.5
ingest.jump_reset_min	30	11.5
ingest.clock_skew_s	300	11.5
ingest.service_area_buffer_km	10	11.5
ingest.batch_max	50	11.3
ingest.body_max_kb	8	11.3
ingest.phone_body_max_kb	16	11.7

Key	Default	Section
ingest.global_rps	200	11.8
sos.dedupe_s	60	19.2
sos.cancel_window_s	30	19.6
sos.escalation_min	0, 2, 5, 10	19.3
sos.device_delay_flag_s	60	19.8
stop.radius_m	50	17.4
stop.arrival_speed_kmh	5	17.4
stop.arrival_dwell_s	15	17.4
stop.departure_dwell_s	10	17.4
stop.departure_speed_kmh	10	17.4
eta.speed_min_kmh	10	17.5
eta.speed_max_kmh	60	17.5
eta.speed_default_kmh	20	17.5
eta.speed_window_min	5	17.5
snapshot.cache_s	3	17.3
qr.rotation_s	30	18.1
boarding.window_before_min	10	18.5
boarding.window_after_min	15	18.5
leave.cutoff_before_trip_min	30	18.7
shift.idle_min	20	15.4
shift.idle_speed_kmh	3	15.4
shift.idle_grace_min	10	15.4
shift.timeout_h	14	15.4
shift.late_start_min	10	15.6
shift.late_start_coordinator_min	20	15.6

Key	Default	Section
trip.generate_days_ahead	7	15.5
route.corridor_m	300	20.1
stop.duplicate_m	100	20.1
stop_request.limit_days	7	20.1
notify.quiet_start	22:00	21.4
notify.quiet_end	06:00	21.4
notify.push_per_hour	6	21.3
notify.group_window_s	120	21.3
notify.retry_min	1, 5, 30, 120	21.2
summary.send_time	08:00	21.9
approach.default_min	5 (range 2 to 15)	16.1
app.poll_foreground_s	10	17.2
web.poll_s	10	17.2
session.idle_min	30 (SEC: 15)	24
otp.ttl_min	10	8
otp.attempts	5	8
otp.lock_min	15	8
doc.expiry_warn_days	30, 7	22
service.hours	05:30 to 22:30	31.9
ops.peak_windows	Set at kickoff	31.9
holidays	College holiday calendar, set each year	15.5

36. Data import formats

All files are UTF-8 CSV with a header row, at most 5 MB, and at most 10,000 rows. Dates are YYYY-MM-DD. Times are HH:MM in campus time. The import runs as a dry run first. It shows each row as passed or failed, with the row number and the reason. Commit is blocked while any row fails, unless the coordinator chooses to commit only the passing rows (default: off).

File	Columns	Rules
users.csv	college_id, full_name, email, role, phone, department, year_of_study, date_of_birth, route_code, boarding_stop_code, bus_number, is_primary	One role per row. college_id and email are unique. email must be a college-domain address, except for role DRV. role is STU, FAC, DRV, TEC, or COR. route_code and boarding_stop_code are required for STU. bus_number is required for FAC and DRV. is_primary is yes or no, for FAC. date_of_birth is required for STU (missing means treated as under 18, section 29.2)
buses.csv	bus_number, registration_number, capacity, insurance_expiry, fitness_expiry, pollution_expiry, permit_expiry	Both numbers are unique. Capacity is 1 to 150
devices.csv	device_id, serial_number, imei, iccid	device_id is HW_ followed by 3 or 4 digits. imei is 15 digits. iccid is 19 or 20 digits. Used for provisioning only (section 16.5)
stops.csv	stop_code, name, latitude, longitude, radius_m	Coordinates inside the service area. radius_m defaults to 50
routes.csv	route_code, name	Unique route_code
route_stops.csv	route_code, stop_code, sequence	Sequence is unique within a route, starting at 1

File	Columns	Rules
timetables.csv	route_code, trip_type, departure_time, weekday_mask	trip_type is MORNING or AFTERNOON. weekday_mask is a list such as MON-FRI or MON,WED,FRI
roster.csv	bus_number, driver_email, valid_from	driver_email must match a DRV user

Rules:

- Cells that start with =, +, -, or @ are rejected in import files, to block formula injection.
- Duplicate keys within a file fail every row that repeats them.
- Imports never change existing records silently. An update to an existing record is shown in the dry run as an update, with the old and new values.

37. Edge cases

Each row is a case the system must handle. Each has a test in section 32.

#	Scenario	Required behaviour
1	Two physical devices carry the same device_id	The second device's token fails (401). Sev 2 alert. Nothing is mixed between the two devices
2	Device reboots during a batch	New boot_id. Unacknowledged samples from the old boot are resent with the old boot_id
3	Device clock is two hours ahead	CLOCK_SKEW recorded. Ordering uses received_at. Display uses recorded_at only when GNSS was valid
4	Cellular down for 40 minutes, then samples flush	Samples older than 10 minutes are BUFFERED and never update live state. Live state keeps the newest valid sample
5	Driver's phone dies, hardware keeps working	Hardware stays active. No user-visible change beyond the phone status

#	Scenario	Required behaviour
6	Hardware and phone both fail	UNAVAILABLE after 120 seconds. Coordinator alerted
7	Tracker moved to another bus without a coordinator update	Positions from the tracker and the driver's phone diverge by more than 500 m for 2 minutes while moving. Technician and coordinator alerted as possible tracker swap
8	Two drivers try to start the same bus	The second start is blocked with BUS_ALREADY_ON_SHIFT, showing who holds the shift
9	Student scans QR on a bus not serving their route	NOT_ON_ROUTE. No record written
10	Student changes route mid-term	New dated assignment. Past attendance unchanged
11	Trip cancelled after some students boarded	Boarded records kept. Trip status CANCELLED. Summary marks it cancelled
12	Driver tries to end shift while SOS is open	Blocked. Only the coordinator can end it, with a reason
13	SOS from a bus with no hardware	Incident created from the phone trigger. Hardware button absent, noted in the incident
14	Coordinator revokes a device while its bus is on shift	Hardware rejected within 5 seconds. Phone continues. Alert raised
15	Student deactivated on a boarding day	Removed from expected lists from the effective date. Existing boardings kept
16	Bus has no faculty assigned	Alerts go to the coordinator. Console shows a warning: no faculty assigned
17	Route has no students	Trip runs normally. Attendance list empty. No absence alerts
18	Route geometry edited during a trip	Trip keeps the geometry version active at its start. Next trip uses the new version

#	Scenario	Required behaviour
19	Stop removed while the bus is heading to it	Bus skips the stop in the ETA list. Driver sees a notice
20	Device flushes 24 hours of buffered data	About 173 requests of 50 samples, sent at one request per 2 seconds, about 6 minutes. Live state updates only from fresh data
21	Hardware and phone are both fresh but 300 m apart	Active source is used. Difference recorded. Alert only if it lasts 10 minutes
22	Phone reports a mock location	Stored and flagged. Never live. Driver warned that the location appears simulated. Coordinator alerted
23	Notification for a deactivated user	Suppressed. Logged as SUPPRESSED
24	Student assigned to two routes for the same trip type	Second assignment blocked (R-06)
25	Export of 120,000 rows	Split by date range. User told which ranges are included
26	Screenshot of a boarding QR code	The code is valid for one 30-second slot only. Residual risk recorded and accepted
27	New SOS from the same bus after the earlier incident was resolved	New incident. Joining applies only to OPEN incidents
28	Bus loses all power during an SOS	Incident sent only if the device had backup power to send. Limit recorded in HW-SPEC
29	Threshold changed during a trip	Applies from the next evaluation. Change logged
30	Public holiday	Trip generator skips dates in the holiday calendar (section 35)

38. Release gate

Production launch needs every item below. Each item is signed off by the named owner in the release record.

- All acceptance criteria (AC) in sections 16 to 19 pass in staging.
- Test suites T-ING, T-SEL, T-SHF, and T-SOS pass.
- Authorisation matrix T-SEC-01 passes.
- Contract tests pass for every firmware version in the field (T-CON).
- Load test T-LOAD-01 and SOS burst test T-LOAD-02 pass.
- Failover drills T-FO-01 to T-FO-04 pass.
- The penetration test has no open high or critical findings.
- Automated accessibility scan (T-A11Y-01) has no critical violations.
- Counsel has signed off the privacy defaults, the consent method, and the limited mode in section 29.
- Retention jobs verified on staging data.
- Runbooks RB-01, RB-02, RB-03, and RB-08 are written and rehearsed.
- A restore drill has been completed in the last 30 days.
- The pilot exit criteria are met (section 39).
- The college has signed off every decision in section 41 marked before launch.

39. Milestones and definition of done

Dates are set at kickoff, once the team size is known. Do not commit dates before then.

Milestone	Scope
M0 Foundation	Repository, environments, CI and CD, SSO and OTP sign-in, initial data model migrations, observability baseline
M1 Ingestion engine	Device and driver endpoints, validation pipeline, storage, simulator, source selector, snapshot API, contract test harness
M2 Student and faculty apps	Home, bus detail, ETA, stop requests, notifications, attendance view
M3 Driver app and phone tracking	Sign-in, pre-trip check, shift, background location, buffering, mode switch, QR display, SOS
M4 Coordinator and technician consoles	Buses, devices, routes, rosters, faculty, imports, SOS console, tickets, firmware rollout

Milestone	Scope
M5 Attendance and alerts	Boarding flows, absence rule, leave, daily summary, operational alerts
M6 Reports and privacy	Reports, data request workflows, parental consent flow, retention jobs
M7 Hardening and pilot	Load tests, failover drills, penetration test, accessibility, two-week pilot on three buses, then staged rollout to the full fleet

Definition of done, applied to every item:

- Code reviewed and merged.
- Tests written and passing (section 32).
- Acceptance criteria verified.
- API documented in the generated OpenAPI file.
- Logs and metrics added (section 31).
- Security checks pass (section 28).
- UI checked for accessibility (section 30).
- Runbook updated if operations change.
- Risky features behind a flag (section 33.5).
- Release notes written.

Pilot exit criteria: live position available for at least 99 percent of active-shift minutes. SOS delivered within 5 seconds at the 95th percentile in drills. At least 95 percent of boardings recorded within 2 minutes of departure. No cross-role exposure found. No Severity 1 incident open.

40. Future features (after MVP)

- RFID check-in, using the hooks in section 18.8.
- Parent accounts with consent: read-only tracking and absence notices.
- Live push (Server-Sent Events or WebSockets with pub/sub) to replace polling.
- Snap-to-route marker display and improved ETA.
- Telephone escalation for SOS tiers 2 and 3.
- Occupancy estimate from boardings and alightings.

- Configurable absence rules per route.
- Predictive ETA using traffic data.
- Automatic delay notices from schedule deviation.
- In-app messages between faculty and drivers.
- Ratings and feedback on buses.
- PDF reports and scheduled report emails.
- Multi-college tenancy in the interface.
- Google Maps layer option.
- Root and jailbreak detection, and certificate pinning, in the app.
- Nearest-stop help for students, using phone location in the foreground only, with consent.
- Vehicle document uploads with expiry tracking.
- Offline mode for the coordinator console.
- Languages beyond English.

41. Open decisions

Each decision needs an owner's sign-off. The recommendation applies unless the owner changes it.

#	Decision	Recommendation	Owner	Needed by
1	Database: PostgreSQL with PostGIS, or MongoDB Atlas (the earlier plan)	PostgreSQL. It supports geographic queries for route corridors and ETA, dated links, and transactions for SOS	Engineering lead	Before M0
2	Cloud provider	AWS, in an India region, using ECS, RDS, and ALB	Engineering lead, college IT	Before M0
3	Map provider	MapLibre with OpenStreetMap, with a layer switch for Google Maps later	Product owner	Before M2
4	Boarding method at launch	QR at launch. RFID in phase 2	Transport office	Before M5

#	Decision	Recommendation	Owner	Needed by
5	Driver phones	College-issued phones for each bus where budget allows. Personal phones with consent as a fallback	Transport office, counsel	Before M3
6	Security staff visibility of all active buses (section 5)	Read-only, all active buses	Security office, counsel	Before M4
7	Parental consent method (section 29.2)	Guardian verification by one-time code, confirmed by counsel	Counsel	Before M6
8	Limited mode for under-18 students without consent (section 29.2)	As specified. Counsel confirms	Counsel	Before M6
9	Retention periods (section 29.4)	As listed. Policy owner confirms	Policy owner	Before M6
10	Emergency escalation contacts and on-duty lists	College provides them	Security office, coordinator	Before M3
11	Critical Alerts for SOS on iOS	Not in MVP. Time-sensitive interruption used	Product owner	Before M3
12	Morning-present, afternoon-absent rule (section 18.6)	As specified. Faculty committee confirms	Faculty committee	Before M5
13	Default mode and mode switching (section 13.2)	AUTO by default. Drivers can switch	Transport office	Before M3
14	GNSS source (module internal or external)	Confirm in HW-SPEC. Software unchanged either way	ECE team	Before firmware freeze

#	Decision	Recommendation	Owner	Needed by
15	Supported OS versions (section 30)	Android 10 and later, iOS 16 and later. Confirm against devices in use	Product owner	Before M2
16	Regional disaster recovery (section 34.12)	Accept the 4-hour regional recovery for MVP, or add a standby region	Super admin, college IT	Before launch

Document control

Version	Date	Change	Author
1.0	18 Oct 2026	First complete draft for build	Product and engineering

Role	Name	Approval	Date
Product owner			
Engineering lead			
ECE lead (hardware interface)			
College counsel (privacy)			
Transport office			